



Unlocking the power of optimized data balancing ratios: a new frontier in tackling imbalanced datasets

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Abstract

Data balancing methods eliminate the problem of imbalanced class distributions, which often lead to the majority class being well-learned while the minority class remains underrepresented, negatively affecting classification performance. This study applies data balancing to the healthcare domain, a critical field where classification success directly impacts human life. The primary aim is to introduce novel balancing methods while addressing the previously overlooked problem of optimizing data balancing ratios. Six healthcare datasets were used: Wisconsin Diagnostic Breast Cancer (WDBC), Wisconsin Prognostic Breast Cancer (WPBC), Z-Ali-zadeh Sani, Kidney, Diabetes, and Stroke, all characterized by significant diseases and imbalanced class distributions. Six balancing methods were tested, including synthetic minority oversampling technique (SMOTE), adaptive synthetic sampling (ADASYN), support vector machine-SMOTE (SVM-SMOTE), Borderline-SMOTE, cubic interpolation, and quadratic interpolation, with interpolation-based methods being adapted to this domain for the first time. The critical factor in data balancing is identifying the optimal ratio that maximizes classification performance. In this study, particle swarm optimization (PSO), whale optimization algorithm (WOA), and Optuna optimization methods were used to optimize balancing ratios via a custom-designed fitness function that simultaneously optimizes classification accuracy and resource consumption. Classification was conducted for three scenarios: full balance, optimized balance, and imbalance, using support vector machine (SVM), random forest (RF), and ensemble learning (EL) classifiers, allowing for extensive analysis. Each combination of balancing methods, classifiers, and optimization techniques was separately analyzed using metrics such as accuracy, precision, recall, F1-score, time, central processing unit (CPU) usage, and memory usage. As a result, the combination that optimally balances classification accuracy and resource consumption was determined for each dataset, providing both comprehensive analysis and insights into the impact of balancing ratio optimization on diagnostic success in health care.

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Keywords Data balancing · Class imbalance · Particle swarm optimization · Support vector machine · Imbalanced datasets

1 Introduction

Class imbalance [1–3] is a situation in a dataset where the number of samples observed between classes varies, and it can significantly affect the performance of classification models. This type of imbalance leads to classification algorithms becoming overly biased toward the majority class and inadequately representing the minority class. Therefore, addressing data imbalance is crucial for enhancing the accuracy and overall effectiveness of classification models. Various data balancing techniques, such as artificial sample generation, oversampling, undersampling, or algorithmic adjustments, can be employed for this purpose. Oversampling methods are often preferred in this field compared to other techniques. Oversampling involves generating new, synthetic samples from existing samples in the minority class or replicating existing samples. This method helps the model learn better about the minority class and improves its ability to recognize samples from this class, thereby reducing bias toward the minority class and increasing the overall classification accuracy.

Class imbalance remains a pivotal challenge in machine learning, necessitating diverse strategies to enhance classification accuracy across various domains. When collecting medical datasets, the distribution between classes is often neglected. Therefore, the imbalance problem appears more intensely in healthcare datasets compared to other fields. Various studies in the literature address this issue. One such study, conducted by Yuan et al. [4], introduced the clustered histogram-based synthetic minority oversampling technique (CHSMOTE), which aims to overcome the limitations of SMOTE by leveraging convex hulls to generate synthetic samples and effectively maintain intra-class relationships, thereby improving classifier performance. Additionally, Yeung et al. [5] suggested that loss functions used in deep learning methods could provide a solution to the class imbalance problem, proposing a loss function called unified local loss. They tested their loss function on five different open datasets (CVC-ClinicDB, Digital Retinal Images for Vessel Extraction (DRIVE), Breast Ultrasound 2017 (BUS2017), Brain Tumour Segmentation 2020 (BraTS20), and Kidney Tumour Segmentation 2019 (KiTS19)) and reported its success in image segmentation. Moradi et al. [6] optimized instance selection using a hesitation-based method, tackling class overlap and imbalance concurrently. Rodríguez-Torres et al. [7] experimented with SMOTE and its derivative FAST_SMOTE on Ecoli and Mammograph datasets, highlighting the importance of data balancing methods in healthcare applications. Addressing health-related challenges, Ishaq et al. [8] improved survival prediction for heart failure patients using SMOTE and effective data mining techniques, and Bellinger et al. [9] introduced SWIM, a framework for extreme imbalance classification that samples with the majority class. Also, Hayashi et al. [10] discussed a one-class ensemble classifier for addressing data imbalance challenges in Ecoli and Mammograph datasets. Sowjanya et al. [11] developed distance-based SMOTE (D-SMOTE) and bi-phasic SMOTE

(BP-SMOTE) methods to reduce dataset imbalance, achieving higher accuracy compared to basic SMOTE by combining these methods with selective classifiers. Subsequently, a stacking ensemble framework was developed, incorporating machine learning, deep learning, and ensemble algorithms. The Framingham dataset was used for data balancing, the Wisconsin Hospital Breast Cancer dataset was utilized for stacked convolutional neural network (CNN), and a dataset predicting COVID-19 cases was employed for stacked recurrent neural network (RNN). Kosolwattana et al. [12] introduced the self-inspected adaptive SMOTE (SASMOTE) method to generate higher-quality synthetic samples for the minority class. SASMOTE improves sample quality through adaptive neighborhood selection and filters uncertain samples that do not differentiate from the majority class via an uncertainty elimination mechanism. The method demonstrated better performance than SMOTE-based methods in healthcare applications such as gene risk discovery and congenital heart disease prediction. Rajalakshmi et al. [13] proposed a novel model combining data balancing, feature selection, and deep learning methods for diabetes prediction in an Internet of Things (IoT)-based healthcare environment. Real-time IoT data collected from the Pima Indian Diabetes dataset were balanced using SMOTE, followed by feature selection using a k-means-based sailfish optimization method. Finally, these features were fed into a bidirectional long short-term memory (LSTM) model, and the proposed system showed superior prediction performance compared to existing methods. Polo et al. [14] conducted a comprehensive evaluation of data balancing methods used for detecting health insurance fraud. Techniques, such as random oversampling, undersampling, SMOTE, and instance hardness threshold, were tested with the XGBoost algorithm. Results showed that SMOTE and instance hardness threshold methods provided consistent results, particularly with a 70% class distribution, and instance hardness threshold exhibited the best performance at a 90% distribution. The study identified important features for fraud detection and contributed to the development of more effective fraud prevention strategies. Chowdhury et al. [15] addressed the challenge of imbalanced data in identifying diabetic cases by employing various data augmentation techniques, including SMOTE-N, edited nearest neighbor (ENN), SMOTE-Tomek, and SMOTE-ENN, to balance the training data before applying machine learning algorithms. Their work highlights the importance of preventing data leakage and provides a comprehensive framework for healthcare practitioners to make informed decisions using machine learning predictions. Elangovan et al. [16] conducted a comprehensive analysis of features to enhance stroke prediction effectiveness using a reliable dataset from the Kaggle website, which has a class imbalance problem. They addressed this issue by employing oversampling techniques like SMOTE and ADASYN to balance the dataset, leading to improved evaluation metrics. Their proposed hybrid neural network and random forest (NN-RF) model, utilizing the balanced dataset, achieved the highest F1-score of 75%, significantly outperforming models trained on the original unbalanced dataset. Mia et al. [17] investigated the application of machine learning algorithms, including logistic regression (LR), RF, and k-nearest neighbor (KNN), to predict cerebral strokes using data from the Harvard Dataverse Repository. They addressed class imbalances with techniques like SMOTE, ADASYN, and ROSE to improve minority class accuracy. Their study introduced a novel hybrid approach

combining ADASYN with RF, achieving a detection high accuracy, demonstrating the effectiveness of this method in stroke prediction. Nizam-Ozogur et al. [18] addressed this challenge by presenting the GASMOTEPSO-ENN method, which combines SMOTE and ENN algorithms using genetic algorithm (GA) and PSO heuristics as a preprocessing method to classify imbalanced health datasets. They evaluated their method on chronic kidney disease (CKD), cerebral stroke prediction (CSP), and PIMA Indian Diabetes (PID) datasets, achieving high-performance metrics. Similarly, Gholampour et al. [19] highlighted the significant challenge posed by dataset imbalances in medical predictive modeling. They investigated the impact of the inherent nature of medical data on the classification of minority classes using ten machine and deep learning classifiers across medical and financial datasets. Despite using techniques like SMOTE, none of the classifiers effectively classified the minority class in the medical dataset, achieving sensitivity below 5.0% and AUC below 57.0%. This contrasted sharply with their success in the financial dataset, where accuracy exceeded 95.0%, sensitivity over 73.0%, and area under the curve (AUC) above 96.0%. Their findings questioned the clinical validity of SMOTE and highlighted the need for advanced modeling techniques that are both theoretically robust and clinically relevant. In another study, Ameksa et al. [20] tackled the issue of stroke prediction by presenting a fusion framework combining multiple base classifiers and a meta-classifier. They compared oversampling, undersampling, and Tomek-link SMOTE (SMOTE-TL) methods to address class imbalance. Their findings revealed that AdaBoost as a meta-classifier achieved the highest performance, with a recall of 88.09% and an F1-score of 83.66%, providing valuable insights into stroke prediction and healthcare interventions. Collectively, these studies represent a comprehensive effort across the spectrum of machine learning to develop robust solutions for the pervasive issue of class imbalance.

Beyond the healthcare domain, various studies have been conducted to address the problem of class imbalance in other fields. This section provides an overview of data balancing methods applied in different domains. In the financial sector, Sun et al. [21] utilized an AdaBoost-SVM ensemble combined with SMOTE and time weighting to dynamically address class imbalances in financial distress predictions. Zhang et al. [22] found the SMOTE method ineffective and proposed KDENS-SMOTE, a modified version of SMOTE based on kernel density and neighbor density. They tested this method on the Moons and Circles dataset and reported improved results compared to the original SMOTE method. Similarly, Rendón et al. [23] highlighted the contribution of SMOTE, SMOTE + ENN, and SMOTE + TL to classification performance on a remote sensing dataset, frequently used in imaging applications. Seiffert et al. [24] addressed class imbalance by combining random undersampling with boosting in the RUSBoost method, achieving success across 15 datasets from different fields. Moreover, Li et al. [25] employed generative adversarial networks (GANs) for fault diagnosis in rotating machinery, significantly improving diagnostic accuracy by generating synthetic samples for new images. Xiao et al. [26] investigated class imbalance in federated learning, emphasizing the need for specialized approaches in decentralized environments. Davagdorj et al. [27] applied SMOTE and ADASYN balancing methods to a smoking cessation raw dataset, analyzing their contributions to this area. Ganganwar et al. [28] focused on synthetic

data generation for sentiment classification, using transformer models to produce synthetic data and achieving promising results. Razavi-Far et al. [29] introduced two novel imputation-based balancing methods, comparing them to classical data balancing methods. Their results indicated that the proposed methods yielded successful outcomes across various fields. In the safety domain, Sangapu et al. [30] analyzed the effectiveness of SMOTE and its derivatives in autonomous vehicle misbehavior detection. Finally, in meteorology, Attaallah et al. [31] developed SMOTE-deep neural networks (DNN) for air pollution forecasting and air quality index (AQI) classification.

A review of the literature reveals that various studies have been conducted to address the problem of dataset imbalance across different domains. However, a detailed examination suggests that the issue of dataset balancing has not received the necessary attention. In the healthcare domain, most studies focus on addressing imbalances specific to a single disease [7, 8, 10–13, 15–17, 20], with little to no evaluation of balancing methods across multiple diseases. Additionally, when examining the balancing techniques used, it is evident that SMOTE and its derivatives dominate the literature, indicating a stagnation in the field. This highlights the need for new methods or adaptations of techniques that have been successful in other domains. Another gap in the literature is the lack of consideration given to balancing ratios. Researchers have not measured the value that different balancing ratios could add to their studies, leading to uncertainty about whether the selected ratios truly yield optimal results. Moreover, the evaluation of existing studies primarily focuses on classification accuracy. However, methods with high resource consumption, regardless of their accuracy, are unlikely to be practical in real-world applications. Considering the massive amount of data available today, focusing solely on classification accuracy is a significant limitation. Without data improvements, it has become nearly impossible to develop effective and widely applicable methods. Therefore, this study aims to provide new perspectives on data balancing as a form of data improvement while enabling researchers to analyze the contributions of these improvements effectively.

Today, data volumes have reached immense levels, further emphasizing the importance of data improvement methods in classification problems. Dataset balancing is one such method, as imbalanced datasets often contain a large number of examples for one class while others remain underrepresented, negatively affecting classification performance, particularly on test data. This issue is especially critical in the healthcare field, where imbalanced datasets can compromise the accuracy of diagnostic models. This study focuses on balancing datasets related to globally prevalent diseases and optimizing the amount of synthetic data generation to enhance classification performance. Distinguishing features of this work include the adaptation of interpolation-based techniques for dataset balancing, their application in this domain for the first time, and being the first study to optimize dataset balancing ratios. Additionally, it uniquely evaluates classification performance by considering both accuracy and resource consumption. Six healthcare datasets—WDBC, WPBC, Z-Alizadeh Sani, Kidney, Diabetes, and Stroke—were used and balanced using traditional methods such as SMOTE, ADASYN, SVM-SMOTE, and Borderline-SMOTE, as well as newly adapted interpolation-based methods. The critical aspect

of the study lies in determining how much synthetic data should be generated for the minority class, an area that has been largely overlooked in the literature. Optimization methods, including PSO, WOA, and Optuna, were employed to simultaneously maximize classification performance and minimize resource consumption. The study evaluated classification performance under different scenarios—imbalance, full balance, and optimized balance—using SVM, EL, and RF classifiers. Results were assessed based on classification accuracy as well as time, CPU usage, and memory usage metrics, thereby determining the real-world applicability of the proposed methods. The contributions of method are like in the below;

1. *Broad Application Across Health-related Datasets* As seen in the literature, existing studies have primarily focused on specific diseases, aiming to facilitate their diagnosis. Unlike these studies, this research conducts analyses on various diseases that are globally prevalent and emphasizes the critical importance of early diagnosis for these conditions.
2. *Pioneering Use of Interpolation-based Balancing Methods* Researchers have often overlooked the importance of the dataset imbalance problem and instead modified and utilized SMOTE and its derivatives, which generate synthetic data linearly. In this study, the contribution of nonlinear synthetic data generation to the field is discussed for the first time. Cubic and quadratic interpolation methods were uniquely adapted to address the dataset balancing problem.
3. *First-time Optimization of Data Balancing Ratios* Dataset balancing is as important as determining the optimal amount of synthetic data generation to enhance classification performance. While some datasets achieve maximum performance with minimal synthetic data, others, particularly those related to certain diseases, require larger amounts of data generation. This issue has not been previously discussed in the literature, and the contribution of balancing ratio optimization to the field has not been evaluated. In this study, solutions to this problem were proposed using optimization methods such as PSO, WOA, and Optuna.
4. *Custom-Designed Objective Function for Optimization Methods* The originality required in optimization methods lies in the design of the fitness function. Our study aims to optimize dataset balancing by simultaneously improving classification accuracy and minimizing resource consumption. To achieve this, a novel fitness function has been specifically designed. Furthermore, by addressing resource consumption, the study is expected to provide researchers with a new perspective in this field.
5. *Comprehensive Evaluation Across Multiple Metrics* A review of existing studies reveals that researchers predominantly focus on classification accuracy. While classification accuracy is crucial in the healthcare domain—where even minor improvements can translate into saving lives—methods with high accuracy but excessive resource consumption and runtime are challenging to implement in real-world applications. Therefore, it is essential to consider resource efficiency alongside classification performance. To address this gap in the literature, this study evaluates each method not only in terms of classification accuracy but also through metrics such as runtime, memory usage, and CPU usage, providing a comprehensive assessment of their practical applicability.

2 Methodology

Class imbalance refers to situations in datasets where the distribution of examples across classes is uneven. This imbalance often results in the majority class being well-learned, while the minority class is barely learned, if at all. Studies conducted on imbalanced datasets may yield good results during the training phase but fail to generalize when exposed to different types of data belonging to the minority class. To address this issue, data balancing methods have been proposed in the literature. However, the importance of dataset balancing as a preprocessing step is often overlooked. A review of the literature reveals two significant gaps: the lack of new data balancing methods and the absence of research on determining the optimal balance ratio that maximizes classification performance. The neglect of these aspects is partly due to the ability of complex classifiers to achieve notable success despite imbalanced data. However, this reliance on complexity introduces new challenges, as increased complexity leads to higher resource consumption, making the methods impractical or even unusable in real-world applications. This study aims to provide a comprehensive analysis of data balancing as a preprocessing step, offering a fresh perspective on the topic while demonstrating that simple classifiers can achieve successful results with balanced datasets.

Health care is one of the domains where data is most abundant and data improvements are most urgently needed. Even the smallest improvement in this field is of great importance as it directly impacts human lives. This study utilizes six datasets characterized by imbalanced class distributions and significant relevance in terms of patient populations: WDBC, WPBC, Z-Alizadeh Sani, Kidney, Diabetes, and Stroke. To address the problem of data imbalance, classic balancing approaches, such as SMOTE, SVM-SMOTE, Borderline-SMOTE, and ADASYN, were applied, alongside the novel adaptation of interpolation-based cubic and quadratic interpolation methods to the data balancing problem for the first time. The aim is to broaden the narrow scope in which the literature has largely been confined. Another limitation observed in prior studies is the use of fixed balancing ratios for data balancing. These fixed ratios cannot guarantee optimized results for every dataset. To address this, the study focuses on optimizing data balancing ratios using optimization techniques such as PSO, WOA, and Optuna. The critical aspect of these optimization methods is the design of the fitness function specific to the problem. In this study, a novel fitness function incorporating both accuracy and resource consumption was designed. The datasets generated after each balancing and optimization step were classified using SVM, RF, and EL classifiers. A total of 54 different combinations were tested, and their performance was evaluated under three scenarios: datasets with optimized balance ratios, datasets with full balance, and imbalanced datasets. Performance metrics included accuracy, precision, recall, F1-score, time, CPU usage, and memory usage, providing insights not only into classification success but also into the resource requirements of each approach. These evaluations identified the most suitable combination for each dataset, offering a comprehensive understanding of both effectiveness and practicality, in contrast to the predominantly accuracy-focused evaluations in the existing literature.

This study serves as a valuable resource for researchers in today's data-driven world, where the volume of data is rapidly increasing, and data analysis has become as important as classification itself. It is evident that the significance of data preprocessing methods will grow in the future, and such studies will be increasingly needed in the literature. Therefore, the analyses conducted in this study are particularly important from this perspective. The workflow of the study is presented in Fig. 1.

The organization of the paper is structured to cover all essential aspects of the research methodology and findings comprehensively. Section 2 details the datasets used, providing a foundation for understanding the data's origin and characteristics. Section 3 discusses the data balancing methods employed, explaining the techniques and their relevance to addressing class imbalance. Section 4 outlines the optimization process, describing how it was used to optimize data balancing ratios. Section 5 describes the classification methodology, detailing how the classifiers were applied to the optimized datasets. Continuing from this structure, Sect. 6 evaluates the performance metrics to assess the effectiveness of the proposed method, analyzing the results across different metrics like accuracy, precision, recall, F1-score, time, memory usage, and CPU usage. Section 7, titled "Evaluation Results," presents a detailed analysis of the experimental results, offering insights into the performance of the proposed method under various scenarios. Section 8, "Discussion," interprets the results, comparing them with existing methods, discussing the implications of the findings, and exploring potential improvements. Finally, Sect. 9, "Conclusion," summarizes the key findings, contributions to the field, and future research directions based on the outcomes of the study. This comprehensive organization ensures a clear flow of information and a thorough exploration of the study's objectives and results.



Fig. 1 The flow diagram of proposed method

2.1 Common characteristics of the datasets

Health care is one of the domains with the highest data density. However, alongside this abundance of data, the class imbalance within datasets is equally significant. This is because healthcare professionals often collect data without considering whether individuals are healthy or ill, resulting in imbalanced datasets classified as disease-related. As a rich source of data, the healthcare domain is particularly appealing to researchers. The primary aim of this study is to develop new balancing methods to address the class imbalance problem in datasets and demonstrate the impact of balancing ratios on dataset balancing. For this reason, datasets related to globally prevalent diseases with imbalanced class distributions have been selected for this study. Experiments were conducted on six healthcare datasets (WDBC, WPBC, Z-Alizadeh Sani, Kidney, Diabetes, and Stroke). These datasets share common characteristics: the diseases they represent are widespread globally, and their data distribution is imbalanced. By selecting datasets associated with common diseases, the proposed methods have the potential to impact large populations. It is expected that studies conducted on these datasets will provide researchers with a new perspective on addressing the challenges of increasing and improving data in today's complex problems.

2.1.1 WDBC dataset

Breast cancer is a common disease among women and one that can be effectively treated if diagnosed early. It is a highly popular research topic among scientists, and the WDBC dataset is frequently used for this purpose. The WDBC dataset [31], developed by Dr. William H. Wolberg at the University of Wisconsin Hospitals in Madison, serves as an essential tool in medical machine learning for breast cancer diagnosis. The dataset was gathered using fine needle aspirate (FNA) of breast masses, a technique that involves extracting cell samples from breast lumps using a thin needle. These samples were then digitized, resulting in the creation of features that describe the characteristics of the cell nuclei visible in the images. The dataset includes 569 cases, classified into 357 benign and 212 malignant cases, reflecting a class imbalance ratio of approximately 0.59 which is calculated using Eq. 1 (minority class size is represented with $N_{\text{minority_samples}}$, and majority class size is represented with $N_{\text{majority_samples}}$). This ratio highlights a significant imbalance, as an ideal balance where the ratio approaches 1 would indicate an equal number of cases in each category, which is optimal for training unbiased machine learning models. In terms of data features, the WDBC dataset consists of 30 real-valued features extracted from the digitized images of the cell nuclei. These features include measurements such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension of each cell nucleus. Each feature is crucial for the diagnostic process, providing comprehensive insights into the cellular structures that may indicate malignancies.

$$\text{Imbalanced Ratio (IR)} = \frac{N_{\text{minority_samples}}}{N_{\text{majority_samples}}} \quad (1)$$

This detailed characterization of cell nuclei and the dataset's intrinsic class imbalance have made the WDBC dataset a popular choice for developing and testing algorithms [32–34] that address not only the accuracy of classification but also the handling of imbalanced data, which mirrors real-world clinical settings. The utilization of WDBC has been featured in numerous scholarly articles, underscoring its value in improving diagnostic algorithms for breast cancer and advancing the field of medical informatics.

2.1.2 WPBC dataset

Another commonly used dataset for breast cancer research is the WPBC dataset. Compared to WDBC, WPBC exhibits a more imbalanced class distribution. This characteristic helps demonstrate the effectiveness of the proposed approach across different breast cancer datasets. The WPBC dataset [35], like the diagnostic counterpart, was also developed at the University of Wisconsin Hospitals in Madison by Dr. William H. Wolberg. This dataset was specifically designed to assess the prognosis of breast cancer, providing a more focused set of data for studies related to the outcomes of breast cancer treatments. The WPBC dataset includes data from 198 patients who had undergone surgery for breast cancer. Among these, there are 151 nonrecurrent cases and 47 recurrent cases, resulting in a class imbalance ratio of approximately 0.31 (minority class divided by majority class, i.e., 47 recurrent divided by 151 nonrecurrent). This significant imbalance poses challenges for machine learning models, as a more balanced ratio closer to 1 would ideally provide a neutral ground for algorithm training, reducing biases toward the majority class. Regarding data features, the WPBC dataset consists of 34 attributes. These include the same 30 real-valued features found in the WDBC dataset, such as measurements of the cell nuclei's radius, texture, perimeter, area, smoothness, and others. Additionally, the WPBC dataset includes four extra features related to the follow-up data: time to recurrence, tumor size, lymph node status, and a binary outcome attribute indicating recurrence or non-recurrence. These prognostic features are particularly vital for models predicting treatment outcomes and assessing long-term survival probabilities. The WPBC dataset's prognostic focus and its inherent class imbalance have been widely discussed and employed in the academic literature [36–38], particularly in studies aimed at improving the predictive accuracy and robustness of prognosis models in oncology. The dataset not only allows researchers to refine diagnostic techniques but also aids in understanding the complexities of cancer progression and treatment efficacy, contributing significantly to advancements in medical informatics and patient care.

2.1.3 Z-Alizadeh Sani dataset

Coronary artery diseases (CAD) are among the most fatal illnesses worldwide. Early diagnosis of this condition can significantly improve individuals' quality of life. For

this study, the Z-Alizadeh Sani dataset was selected, as it is widely used for CAD diagnosis and exhibits a significantly imbalanced class distribution. These characteristics make it highly relevant for demonstrating the effectiveness of the proposed approach in achieving the study's objectives. The Z-Alizadeh Sani dataset [39] is an important resource in the field of cardiovascular research, specifically designed to aid in the diagnosis of CAD. Developed by Dr. Alireza Alizadeh Sani at the Rajaie Cardiovascular Medical and Research Center in Tehran, Iran, this dataset encompasses a diverse patient population, reflecting the complexity of CAD diagnosis. The dataset comprises data from 303 patients, divided into two primary classes: those diagnosed with CAD and those without. Among these, 216 patients were diagnosed with CAD and 87 were categorized as not having CAD, resulting in a class imbalance ratio of approximately 0.40.

This imbalance is significant and reflects the challenges inherent in medical datasets where certain conditions, such as CAD, are more prevalent due to the selected patient groups. In terms of features, the Z-Alizadeh Sani dataset contains 54 features, which include basic demographic information, blood pressure readings, blood tests results, heart rate, and results from various clinical exams like electrocardiogram (ECG) tests and echocardiogram. It also incorporates risk factors such as smoking status, obesity, family history of heart disease, and patient symptoms. These comprehensive features are instrumental for building predictive models that can accurately differentiate between patients with and without CAD. The rich set of features and the dataset's focus on CAD make it a valuable resource for developing and testing machine learning models aimed at enhancing the diagnostic processes for cardiovascular diseases. The class imbalance and the diversity of medical attributes also pose unique challenges that stimulate research in effective machine learning strategies, especially in balancing techniques to mitigate bias toward the more frequent class. This dataset has been widely used in scholarly articles [40–42] to explore new methodologies in predictive analytics within the healthcare domain, contributing to better clinical decision-making and patient management strategies.

2.1.4 Kidney disease dataset

Kidney diseases are also common among individuals. However, finding datasets with imbalanced class distributions for this condition is quite challenging. In this study, one of the rare imbalanced datasets in this domain, the Kidney Disease dataset, was utilized. The Kidney Disease dataset [43] is an invaluable resource for medical research, particularly in the field of nephrology, focused on the detection and management of CKD. While specific details about its origin are not widely published, it is understood that the dataset aggregates patient data from multiple hospitals and clinics, capturing a broad spectrum of symptoms, test results, and demographic information relevant to kidney health. The dataset comprises records from 400 patients, classified into two distinct groups: those diagnosed with CKD and those without the condition. It includes 250 cases labeled as CKD and 150 cases labeled as no CKD, resulting in a class imbalance ratio of approximately 0.60 (minority class divided by majority class, i.e., 150 non-CKD divided by 250 CKD). This ratio indicates a substantial class imbalance, which

is a common challenge in medical datasets, particularly those involving diseases with variable prevalence. The Kidney Disease dataset is extensively equipped with 24 attributes, including age, blood pressure, specific gravity, albumin, sugar, red blood cells, pus cell, pus cell clumps, bacteria, blood glucose random, blood urea, serum creatinine, sodium, potassium, hemoglobin, packed cell volume, white blood cell count, red blood cell count, hypertension, diabetes mellitus, coronary artery disease, appetite, pedal edema, and anemia. Each attribute plays a critical role in diagnosing CKD, making this dataset a crucial tool for developing predictive models that can effectively identify early stages of the disease. Due to the dataset's comprehensive nature and the inherent challenges posed by class imbalance, it is a popular choice for research [44–46] aimed at improving diagnostic algorithms in nephrology. Researchers frequently utilize this dataset to test new machine learning methods, particularly those that address the imbalance issue, thereby advancing the capabilities of medical diagnostic tools and improving patient outcomes in CKD management.

2.1.5 Diabetes dataset

Diabetes is a critical disease that can affect individuals regardless of age. Early diagnosis of this condition is highly important and has garnered significant interest among researchers. In this study, the Diabetes dataset, known for its imbalanced class distribution, was utilized. The Diabetes dataset [47], particularly renowned as the Pima Indians Diabetes dataset, is a critical tool in medical research for enhancing the diagnostics of diabetes. Originally compiled by the National Institute of Diabetes and Digestive and Kidney Diseases, it consists of data from 768 female patients of Pima Indian heritage, who are part of a community near Phoenix, Arizona, USA. This group is significantly studied due to its higher predisposition to diabetes compared to other populations, making the dataset highly relevant for public health interventions. The dataset is divided into two categories: 268 patients diagnosed with diabetes and 500 patients classified as nondiabetic, yielding a class imbalance ratio of approximately 0.536 (diabetic cases divided by nondiabetic cases). This ratio underscores the prevalent issue of class imbalance that often complicates the training and effectiveness of predictive models in medical machine learning. Comprising eight diagnostic features including the number of pregnancies, plasma glucose concentration, diastolic blood pressure, triceps skinfold thickness, 2-h serum insulin, body mass index, diabetes pedigree function, and age, the dataset provides a comprehensive basis for building models that predict the onset of diabetes. Each feature is chosen for its clinical relevance in predicting the likelihood of diabetes onset, making this dataset a cornerstone for developing advanced diagnostic tools. Given its detailed focus on a high-risk group and the challenge presented by its class imbalance, the PID dataset is extensively utilized in scholarly research [48–51] to develop and refine algorithms that not only improve diagnostic accuracy but also address the biases introduced by uneven class distributions. This research contributes to better understanding and managing diabetes, particularly in genetically predisposed populations.

2.1.6 Stroke dataset

Stroke is a leading cause of death and disability, particularly among elderly individuals. One of the primary objectives of our study is to enable early diagnosis. To align with this goal, the Stroke dataset was selected for the study. The Stroke dataset [52] is a pivotal tool in medical research, enhancing the diagnostics of stroke. Compiled by healthcare professionals, it includes data from 5,110 patients, providing comprehensive insights into stroke risk factors and outcomes. The dataset is divided into two categories: 249 patients diagnosed with a stroke and 4,861 patients classified as non-stroke, resulting in a significant class imbalance ratio of approximately 0.051 (stroke cases divided by non-stroke cases). This imbalance presents challenges in training effective predictive models in medical machine learning.

The dataset comprises ten diagnostic features, including age, hypertension, heart disease, ever married, work type, residence type, average glucose level, body mass index, smoking status, and gender. Each feature is selected for its clinical relevance in predicting the likelihood of stroke, offering a robust basis for developing models that predict stroke onset. The detailed focus on a diverse population and the challenge of class imbalance make the Stroke dataset highly relevant for public health interventions and scholarly research. The dataset's significant class imbalance complicates the development of predictive models, often leading to biased results if not properly addressed. However, it also provides an opportunity to refine algorithms that can effectively manage these biases, improving diagnostic accuracy. Researchers utilize advanced techniques, such as oversampling, undersampling, and synthetic data generation, to mitigate the effects of class imbalance and enhance model performance. The Stroke dataset's comprehensive nature and the high-risk factors included make it an invaluable resource for developing and testing diagnostic tools. These tools are crucial for early detection and intervention, ultimately contributing to better management and prevention of stroke, particularly in high-risk populations.

3 Data balancing methods

Data balancing [53, 54] is a pivotal technique in machine learning, especially crucial in addressing classification problems with significant class imbalances. Imbalanced datasets, where one class is underrepresented relative to others, can lead to biased models that perform poorly on the minority class, often resulting in misclassification. This issue is particularly salient in medical datasets, where accurate identification of rare conditions (e.g., malignant tumors, stroke events) is essential for effective diagnosis and treatment. Implementing robust data balancing strategies allows for the development of more reliable and equitable predictive models. This study aims to explore the contribution of interpolation-based techniques, which are not currently found in the literature, to data balancing, as well as the impact of determining the balancing ratio—often overlooked by researchers—on classification performance and resource consumption. Today, researchers predominantly focus on classifiers, often neglecting or repeatedly using similar data improvement methods. Therefore, the literature requires new approaches that have proven successful in

other domains but have not yet been adapted to this field. Additionally, researchers tend to use fixed balancing ratios without discussing their impact on classification performance. This study seeks to address both issues by offering solutions that contribute to advancing the literature.

In this study, both traditional methods—such as SMOTE, ADASYN, SVM-SMOTE, and Borderline-SMOTE—and interpolation-based methods, specifically cubic and quadratic interpolation, are utilized for data balancing. The fundamental difference between these two types of methods is that while SMOTE and its derivatives generate data linearly, interpolation-based methods produce data nonlinearly. Linear methods, compared to nonlinear ones, have a higher likelihood of introducing noise during data generation. Therefore, there is a clear need for approaches in data balancing that are resource-efficient and capable of nonlinear generation. Another distinguishing feature of this study is the discussion of the impact of optimized balancing ratios on classification performance, as opposed to using fixed balancing ratios. The balancing ratio quantitatively determines the minority class size in the dataset. Generating less data can sometimes hinder classification accuracy, whereas generating more data can increase resource consumption. To address this, optimization methods, such as PSO, Optuna, and WOA, are employed to determine the optimal balancing ratio. A novel fitness function has been designed for these optimization methods, capable of optimizing both accuracy and resource consumption. The balancing ratio is set to vary between 0.9 and 1.1. This range effectively addresses class imbalance while preserving the natural distribution of the dataset, providing a balanced approach. It ensures that the minority-to-majority class ratio is nearly equal, maintaining balance without exceeding a strict 1.0 ratio, thus preserving natural variation. This approach avoids the risk of model overfitting that could arise from overbalancing. Additionally, this range minimizes resource usage and processing time for data augmentation or reduction, reducing the cost of the optimization process, which aligns with the objectives of this study. In the following section, the details of each data balancing method will be presented.

3.1 Data balancing using SMOTE

SMOTE [55–57] is a widely used technique that generates synthetic samples for the minority class by interpolating between existing minority instances. This method offers several advantages. Firstly, by generating synthetic examples, SMOTE enhances the classifier's ability to learn decision boundaries for the minority class, leading to better generalization. Secondly, it reduces the likelihood of overfitting to the minority class training instances, as it creates new, synthetic samples rather than simply replicating existing ones. Additionally, SMOTE is flexible, allowing for adjustments to different levels of interpolation and can be combined with other balancing techniques to suit specific needs.

However, SMOTE also has some disadvantages. One of the main risks is the potential introduction of noise by generating samples that overlap with the majority class, which can lead to misclassifications and blurred class boundaries. Furthermore, it requires careful tuning of parameters such as the number of nearest

neighbors (k) to be effective, which can be complex and time-consuming. Lastly, SMOTE is primarily designed for continuous data, making its application to categorical data challenging and often necessitating additional preprocessing steps. Overall, while SMOTE is a powerful tool for addressing class imbalance, it must be applied thoughtfully to avoid introducing new issues. The step-by-step application of SMOTE like in the below;

1. *Identify Nearest Neighbors* For each minority class instance x_i , its k -nearest neighbors within the minority class are identified using a distance metric, typically Euclidean distance.
2. *Select Random Neighbor* One of the k -nearest neighbors of x_i^k is randomly selected. This step introduces variability and ensures that synthetic samples are spread out across the feature space.
3. *Generate Synthetic Instance* A synthetic instance (x_{new}) is created by interpolating between x_i and x_i^k . The interpolation is done using Eq. 2.

$$x_{new} = x_i + \beta \cdot (x_i^k - x_i) \quad (2)$$

In Eq. 2, β is a random number between 0 and 1.

SMOTE has been chosen for this study due to its well-documented effectiveness in generating synthetic samples for the minority class. By leveraging linear interpolation between existing minority class instances, SMOTE generates new, plausible samples that enhance the learning capability of classifiers. This method is particularly advantageous as it not only addresses class imbalances effectively but also avoids overfitting, a common challenge with simple duplication-based oversampling techniques. In this study, SMOTE plays a pivotal role in creating a more balanced dataset, which is essential for improving classification performance. Two key parameters are critical to its successful application: the value of k (number of nearest neighbors) and the balancing ratio. These parameters directly influence the quality and distribution of synthetic samples and, consequently, the model's ability to generalize effectively. To ensure the optimal values for these parameters, we employ advanced optimization techniques such as PSO, WOA, and Optuna. These methods are used to dynamically determine the best combination of k and balancing ratio, striking a balance between dataset quality, computational efficiency, and model accuracy. PSO is utilized for its ability to explore and exploit the search space efficiently, often leading to globally optimal solutions in parameter tuning. WOA, inspired by the hunting behavior of humpback whales, is applied for its robustness in solving complex optimization problems. Optuna, a cutting-edge hyperparameter optimization framework, is particularly valuable for fine-tuning and identifying the best configurations through efficient sampling methods.

The integration of these optimization techniques with SMOTE not only highlights the practical application and relevance of this method but also underscores its role in achieving the study's overarching goal: finding the optimum data balancing ratio in a cost-effective manner. This approach ensures that the generated synthetic samples contribute to improved model performance while maintaining computational

and resource efficiency. The insights gained from this work further demonstrate the adaptability and effectiveness of SMOTE when paired with advanced optimization techniques, paving the way for its broader application in imbalanced data scenarios.

3.2 Data balancing using ADASYN

ADASYN [58–60] is an advanced oversampling technique that improves upon SMOTE by focusing on generating synthetic samples for minority class instances that are harder to classify. This technique dynamically adjusts the generation of synthetic data based on the learning difficulty of each minority instance, ensuring that more synthetic data is generated for instances that are more challenging to classify. This approach helps in reducing the bias toward the majority class and enhances the classifier's ability to learn more complex decision boundaries. The key advantages of ADASYN include its ability to improve the classifier's performance on difficult-to-classify minority instances and its adaptive nature, which prioritizes instances that need more attention. However, ADASYN also has some disadvantages, such as increased computational complexity due to the additional steps involved in calculating the difficulty levels and generating synthetic instances accordingly. Additionally, like SMOTE, ADASYN can introduce noise if not properly tuned, potentially leading to overlapping classes and misclassification. Despite these challenges, ADASYN is crucial for creating balanced datasets that can lead to more accurate and robust predictive models, especially in medical domains where correctly identifying minority class instances is vital. The step-by-step application of ADASYN like in the below;

1. *Calculate Difficulty Level* The difficulty level d_i for each minority class instance x_i is calculated by determining the number of majority class neighbors within its k -nearest neighbors, according to Eq. 3.

$$d_i = \frac{\text{Number of Majority Class Neighbors}}{k} \quad (3)$$

2. *Determine Number of Synthetic Samples* The total number of synthetic samples G needed for the entire dataset is computed first. Subsequently, the number of synthetic samples G_i to be generated for each minority instance x_i is determined based on their difficulty levels. The calculation of G is given in Eq. 4.

$$G_i = \frac{d_i}{\sum_{i=1}^k d_i} xG \quad (4)$$

3. *Generate Synthetic Instances* For each minority instance x_i , G_i synthetic instances are generated by interpolating between x_i and its k -nearest neighbors, using Eq. 2 for the interpolation process.

ADASYN is selected for this study due to its unique focus on generating synthetic data for minority class instances that are harder to classify, effectively reducing bias toward the majority class. Unlike static oversampling methods, ADASYN

dynamically adjusts the generation of synthetic samples based on the difficulty of learning for each minority instance, prioritizing those farther from the decision boundary or in complex regions of the feature space. This approach enhances the classifier's ability to learn from challenging cases, leading to improved performance on difficult-to-classify instances and ensuring a more balanced and accurate classification outcome.

In this study, ADASYN's application is tailored through the optimization of two critical parameters: k (the number of nearest neighbors) and the balancing ratio. These parameters directly influence the quality and distribution of synthetic samples. To determine their optimal values, advanced optimization techniques—PSO, WOA, and Optuna—are employed. PSO effectively explores the parameter space to identify global optima, WOA contributes robustness in handling nonlinear optimization landscapes, and Optuna efficiently fine-tunes hyperparameters through dynamic sampling. By leveraging these optimization methods, ADASYN is applied in the most effective way, ensuring that the synthetic data generation process maximizes classification performance while minimizing resource consumption. This approach aligns seamlessly with the study's objective of optimizing data balancing ratios in a cost-effective manner, demonstrating how ADASYN, coupled with robust optimization techniques, significantly contributes to achieving accurate and balanced classification in imbalanced datasets.

3.3 Data balancing using SVM-SMOTE

SVM-SMOTE [61–63] is an enhanced version of the traditional SMOTE technique, specifically designed to address some of the challenges associated with generating synthetic samples for the minority class. By leveraging the decision boundary determined by the support vector machine (SVM), SVM-SMOTE focuses on generating synthetic samples in regions where they are most impactful, such as near the minority class boundary. This approach offers several advantages over traditional SMOTE. Firstly, by utilizing the SVM decision boundary, SVM-SMOTE ensures that synthetic samples are generated in areas that contribute to more effective decision-making, reducing the risk of generating noise or overlapping samples in irrelevant regions. This leads to improved class separability and enhances the model's ability to generalize. Secondly, SVM-SMOTE provides better control over the placement of synthetic samples by focusing on "support vectors" and regions closer to the boundary, which are critical for classification performance. Lastly, the method retains the flexibility of SMOTE in terms of interpolation adjustments and can be combined with other balancing strategies for tailored solutions.

However, SVM-SMOTE also has limitations. The reliance on SVM models introduces computational complexity, particularly for large datasets, as the training of SVMs can be resource-intensive. Additionally, selecting appropriate parameters for the SVM model, such as the kernel type and hyperparameters, requires careful tuning to ensure the effectiveness of the synthetic sample generation process. Similar to SMOTE, the effectiveness of SVM-SMOTE depends on selecting the appropriate

number of neighbors (k) and balancing ratio, making optimization crucial. The step-by-step application of SVM-SMOTE is as follows:

1. *Train SVM Model* The SVM model is trained on the imbalanced dataset to determine the decision boundary between classes.
2. *Identify Support Vectors* Support vectors (key instances that define the boundary) of the minority class are identified.
3. *Nearest Neighbor Selection* For each minority support vector x_i , k -nearest neighbors are identified within the minority class using a distance metric.
4. *Generate Synthetic Instance* A synthetic sample (x_{new}) is generated by interpolating between x_i and one of its k -nearest neighbors (x_i^k), using Eq. 5.

$$x_{new} = x_i + \beta \cdot (x_i^k - x_i) \quad (5)$$

In Eq. 5, β is a random value between 0 and 1, ensuring variability in synthetic sample generation. SVM-SMOTE was selected for this study due to its capability to enhance classifier performance by focusing on critical regions of the feature space. Unlike traditional SMOTE, SVM-SMOTE leverages the decision boundary derived from an SVM to generate synthetic samples closer to the class boundary, ensuring a more balanced and well-defined dataset. This targeted approach addresses class imbalance more effectively, particularly in complex classification tasks, by prioritizing the generation of samples that contribute directly to improving the model's learning process. In this study, two key parameters for SVM-SMOTE are optimized: k (the number of nearest neighbors) and the balancing ratio, both of which play a critical role in defining the quality and distribution of synthetic samples. To determine the optimal values for these parameters, advanced optimization techniques—PSO, WOA, and Optuna—are employed. PSO excels in identifying global optima through efficient exploration, WOA provides robust handling of nonlinear optimization problems, and Optuna dynamically fine-tunes hyperparameters for enhanced performance. By combining these optimization methods with SVM-SMOTE, the study ensures that the synthetic samples generated are not only effective in balancing the dataset but also computationally efficient. This integration highlights a robust and cost-effective approach to addressing class imbalance, demonstrating that SVM-SMOTE, when optimized using advanced techniques, contributes significantly to improving classification performance while maintaining resource efficiency. This approach provides a practical solution for imbalanced datasets, making it highly suitable for applications requiring both accuracy and computational feasibility.

3.4 Data balancing using borderline-SMOTE

Borderline-SMOTE [64–66] is an advanced variant of the traditional SMOTE technique, specifically designed to address the challenges of class imbalance by focusing on the decision boundary between classes. Unlike standard SMOTE, which generates synthetic samples uniformly across the minority class, Borderline-SMOTE concentrates on instances near the decision boundary where misclassification risk

is highest. This targeted approach enhances the classifier's ability to distinguish between classes effectively.

One of the key advantages of Borderline-SMOTE is its ability to focus sample generation on critical areas. By identifying "borderline" instances of the minority class—those close to the majority class—it ensures that synthetic samples are generated where they will most effectively enhance classification performance. This approach refines decision boundaries, leading to better generalization and reduced misclassification near class overlaps. Additionally, by avoiding synthetic sample generation in regions far from the decision boundary, Borderline-SMOTE minimizes the risk of introducing noise into the dataset. However, there are some disadvantages to Borderline-SMOTE. The method requires additional computational effort to identify borderline instances, making it more resource-intensive than standard SMOTE. Moreover, its effectiveness depends on tuning parameters such as the number of nearest neighbors (k) and the balancing ratio, which can be time-consuming and complex. Despite its focused approach, there is still a risk of generating synthetic samples that overlap with the majority class if the decision boundary is not well-defined.

The application of Borderline-SMOTE involves several steps. First, minority class instances are classified as "safe," "borderline," or "noise" based on their proximity to majority class neighbors. Only instances identified as borderline are used for generating synthetic samples. For each borderline instance (x_i), its k -nearest neighbors within the minority class are identified. Then, synthetic samples (x_{new}) are created by interpolating between x_i and a randomly selected neighbor (x_i^k), using Eq. 6.

$$x_{new} = x_i + \beta \cdot (x_i^k - x_i) \quad (6)$$

In Eq. 6, β is a random value between 0 and 1, ensuring variability in the generated samples.

Borderline-SMOTE was selected for this study due to its focused approach in generating synthetic samples for instances near decision boundaries, which are critical regions prone to misclassification. Unlike traditional SMOTE, Borderline-SMOTE specifically targets borderline examples, particularly in datasets with overlapping class distributions, ensuring that the classifier can better distinguish between classes in challenging areas. This targeted sampling strategy makes it particularly effective for addressing class imbalance in complex datasets. In this study, two important parameters of Borderline-SMOTE are optimized: k (the number of nearest neighbors) and the balancing ratio. These parameters are crucial for controlling the quality and placement of synthetic samples. Advanced optimization techniques, including PSO, WOA, and Optuna, are employed to determine their optimal values. PSO provides efficient exploration and convergence, WOA ensures robust performance across nonlinear parameter spaces, and Optuna dynamically adjusts hyperparameters to fine-tune the process. This comprehensive optimization framework ensures that the synthetic samples generated by Borderline-SMOTE effectively balance the dataset while minimizing the risk of introducing noise or overfitting. By integrating Borderline-SMOTE with these optimization methods, the study presents

a robust and computationally efficient approach to tackling class imbalance. This method enhances the classifier's performance by focusing on the most challenging areas of the feature space, making it an ideal choice for imbalanced datasets where precision near decision boundaries is critical. The optimized application of Borderline-SMOTE underscores its practicality and effectiveness in building balanced datasets for improved classification outcomes.

3.5 Cubic interpolation for data balancing

Interpolation can be explained as the estimation of unknown data points based on known ones. Interpolation methods are widely used due to their simplicity and effectiveness, particularly in increasing image resolution. In this study, the impact of interpolation, which has proven effective in image processing, is analyzed for its application to data balancing. Typically, SMOTE and its derivatives are used for data balancing. However, these methods face challenges due to their reliance on k-nearest neighbors, making the selection of the k value difficult. Additionally, their linear nature limits their ability to represent complex data, as real-world data is often not linearly distributed. This issue can be addressed by employing nonlinear interpolation techniques. In this study, cubic interpolation is the first interpolation method applied to data balancing. Cubic interpolation [67, 68], applied here for the first time in this context, is a technique that models relationships between data points more accurately and smoothly compared to linear methods. By representing transitions with natural curves, it provides high precision, particularly in graphical applications and physical simulations. Its ability to account for trends between data points reduces errors caused by sudden changes, thereby enhancing the accuracy of predictions. Cubic interpolation ensures derivative continuity by utilizing multiple data points rather than just two, resulting in smoother and more realistic outcomes. This makes it a preferred method in fields such as image processing, time series analysis, and the simulation of physical phenomena, where it reduces noise and better represents the curved structure of the data. Moreover, cubic interpolation has advantages over SMOTE and its derivatives as it does not require parameter tuning. This characteristic makes it a powerful tool in various applications. Additionally, cubic interpolation can mitigate some of the risks associated with overfitting and class overlap that are often observed in linear methods. In this study, cubic interpolation has been adapted for the data balancing problem. The step-by-step application of the modified cubic interpolation method is as follows:

1. *Identifying the Minority Class* Initially, the dataset is divided into two classes: the minority class (C_{min}) and the majority class (C_{max}).
2. *Sorting the Minority Class* The vectors in the minority class are then sorted in ascending order based on their absolute values. This approach ensures that new vectors are generated from similar neighboring vectors. Otherwise, if points with significantly small and large absolute values are treated as neighbors, it may result in noisy data within the dataset. Each vector in the minority class is defined

as $v_i = [x_i, y_i, z_i, \dots]$. The absolute value of these vectors is calculated using Eq. 7, and the sorted sequence of vectors is determined using Eq. 8.

$$|v_i| = \sqrt{x_i^2 + y_i^2 + z_i^2 + \dots} \quad (7)$$

$$v_1 < v_2 < \dots < v_n \quad (8)$$

In Eq. 8, n represents the number of elements in the minority class

3. *Cubic Interpolation Coefficient Calculation* The cubic interpolation coefficients are calculated separately for each component of the vector v . First, two consecutive vectors, (v_i, v_{i+1}) , are selected. Then, the coefficients a, b, c, d for each x, y, z component are computed using Eqs. 9–12, respectively.

$$a = v_i \quad (9)$$

$$b = f'(v_i) \quad (10)$$

$$c = 3x(v_{i+1} - v_i) - 2xf'(v_i) - f'(v_{i+1}) \quad (11)$$

$$d = 2x(v_i - v_{i+1}) + f'(v_i) + f'(v_{i+1}) \quad (12)$$

4. *Determining the Number of Intermediate Samples* After calculating the coefficients, t synthetic data points are generated between each pair of vectors. The value of t is determined using the specified balance ratio (R). This ensures that an equal number of intermediate vectors is generated between each pair of vectors. The t parameter is calculated using Eq. 13.

$$t = \frac{1}{k+1}, \frac{2}{k+1}, \dots, \frac{k}{k+1} \quad (13)$$

In Eq. 13, k represents the number of synthetic samples to be added between two vectors. The value of k is determined using Eqs. 14 and 15.

$$n_{\text{minority}} = |C_{\text{min}}| \quad (14)$$

$$k = \frac{N_{\text{synthetic}}}{n_{\text{minority}} - 1} \quad (15)$$

In the equations, $N_{\text{synthetic}}$ denotes the number of synthetic samples to be generated, while n_{minority} represents the number of examples in the minority class.

5. Generating synthetic samples once the number of synthetic samples to be generated between each pair of vectors is determined, the production of synthetic data begins. The synthetic samples are calculated using Eqs. 16–18.

$$x_{new}(t) = a_x + b_x \cdot t + c_x \cdot t^2 + d_x \cdot t^3 \quad (16)$$

$$y_{new}(t) = a_y + b_y \cdot t + c_y \cdot t^2 + d_y \cdot t^3 \quad (17)$$

$$v_{new}(t) = [x_{new}(t), y_{new}(t)] \quad (18)$$

In the equations, x_{new} and y_{new} represent the newly generated synthetic data points, while v_{new} denotes the balanced dataset.

6. *Creating the Balanced Dataset* After generating new data points for the minority class and appending them to the existing minority vectors, the updated minority vectors are combined with the majority vectors to form the balanced dataset.

Cubic interpolation, despite being an older technique, is adapted for data balancing in this study to showcase its novel application in this domain. Unlike linear methods, cubic interpolation allows for the generation of smoother and more realistic synthetic samples, capturing complex relationships between features more effectively. Cubic interpolation differs from SMOTE and ADASYN in that it primarily focuses on generating high-quality synthetic samples through the fitting of a cubic polynomial to the data points, which inherently captures more complex relationships between features. While SMOTE and ADASYN require the optimization of both the number of nearest neighbors (k) and the balancing ratio, cubic interpolation requires the optimization of only the imbalanced ratio. This simplification allows cubic interpolation to concentrate on achieving the most effective balance between the minority and majority classes, ensuring that the synthetic data generated maintains the nuanced patterns present in the minority class without the additional complexity of tuning multiple parameters. This method is particularly suitable for our aim of optimizing the data balancing ratio in a cost-effective manner because it provides high-quality synthetic data with potentially lower computational costs compared to more complex deep learning approaches.

3.6 Quadratic interpolation for data balancing

In this study, quadratic interpolation is another nonlinear interpolation method applied to data balancing, marking its first use in this context. Quadratic interpolation differs from cubic interpolation in several ways. While cubic interpolation is more complex and slower, quadratic interpolation is simpler and faster. This method is capable of generating synthetic data points that closely resemble the original data. The motivation for using this method is to compare the contributions of simple and complex interpolation techniques to dataset balancing.

Unlike cubic interpolation, which relies on four coefficients, quadratic interpolation uses only three coefficients, making it a second-degree polynomial. The absence of derivative calculations allows quadratic interpolation to be faster than cubic interpolation. Like cubic interpolation, quadratic interpolation [69, 70] generates synthetic samples by fitting a polynomial to the data points, but it uses a simpler quadratic polynomial. This approach provides smoother transitions and more realistic synthetic samples compared to linear methods like SMOTE and ADASYN. Quadratic interpolation is particularly advantageous when the relationship between features is complex and nonlinear. Its smoother and more flexible nature helps capture the underlying structure of the data more effectively, reducing the risk of introducing artifacts or unrealistic samples. Additionally, quadratic interpolation inherently reduces the risk of overfitting, as it represents the data using a simpler model with a lower-degree polynomial. Due to these advantages and its success in applications like image resolution enhancement, this method is expected to perform well in dataset balancing. The step-by-step calculation of quadratic interpolation is as follows:

1. *Dividing the Dataset* Quadratic interpolation uses parabolic curves between two data points to generate synthetic examples while balancing the minority class. This method is both fast and suitable for producing generalizable intermediate points that can help prevent overfitting. Initially, the dataset is divided into minority (C_{min}) and majority (C_{max}) classes.
2. *Mevcut ve Hedef Denge Oranı Belirlenmesi* Mevcutta veri setinin ne kadar dengesiz olduğu (R_{source}) ve dengelemede hedefin ne olduğu belirlenir. Bu çalışmada hedef değer (R_{target}) optimizasyon yöntemleri kullanılarak belirlenmektedir. *Determining the Current and Target Balancing Ratios* The current imbalance ratio of the dataset (R_{source}) and the target balancing ratio (R_{target}) are identified. In this study, the target ratio (R_{target}) is determined using optimization methods.
3. *Calculating the Required Number of Synthetic Examples* Using the target number of examples, the number of synthetic examples to be added to the minority class is calculated using Eq. 19.

$$N_{synthetic} = R_{target} \times |C_{max}| - |C_{min}| \quad (19)$$

In Eq. 19, the required number of synthetic examples is represented by $N_{synthetic}$.

4. *Preparing Data Points for Interpolation* Initially, the absolute values of the minority class vectors are calculated and sorted in ascending order. This minimizes the risk of generating noisy data. Subsequently, two consecutive vectors v_i, v_{i+1} are selected for synthetic data generation.
5. *Defining the Quadratic Polynomial* First, the number of synthetic examples (k) to be generated between two vectors is calculated using Eq. 20. To add k synthetic data points, the quadratic polynomial is defined as shown in Eq. 20.

$$P(t) = a + b.t + c.t^2 \quad (20)$$

In Eq. 20, t represents a value within the range $[0, 1]$, $P(t)$ denotes the polynomial used to generate the vector at intermediate points, and a , b , and c are the coefficients of the polynomial.

6. *Calculating Polynomial Coefficients* To compute the new synthetic vectors, the polynomial coefficients must first be determined. These coefficients are calculated using Eqs. 21–23.

$$a = v_i \quad (21)$$

$$b = v_{i+1} - v_i \quad (22)$$

$$c = v_{i+1} - v_i - b \quad (23)$$

7. *Determining the Value of t* The parameter t is used during the interpolation process to determine the positions of intermediate points along the interpolation curve between two data points (v_i, v_{i+1}) . It takes a value within the range $[0, 1]$ and specifies the location of the intermediate points on the curve. This value is calculated using Eq. 24.

$$t = \frac{1}{k+1}, \frac{2}{k+1}, \dots, \frac{k}{k+1} \quad (24)$$

If the number of intermediate vectors is calculated as 3, the t values will be $[0.25, 0.50, 0.75]$.

8. *Generating Synthetic Examples* Once all coefficients have been determined, the new intermediate points are calculated using Eqs. 25 and 26.

$$x_{new}(t) = a_x + b_x \cdot t + c_x \cdot t^2 \quad (25)$$

$$y_{new}(t) = a_y + b_y \cdot t + c_y \cdot t^2 \quad (26)$$

In the equations, x_{new} and y_{new} represent the newly generated synthetic vectors created between two existing vectors. Finally, the newly generated minority vectors are combined with the majority vectors to form the balanced dataset.

Quadratic interpolation is similarly adapted for data balancing to highlight its novel application. Quadratic interpolation offers smoother transitions and more realistic synthetic data compared to linear methods, making it effective in capturing the underlying structure of the data. Quadratic interpolation differs from SMOTE and ADASYN in that it primarily focuses on generating high-quality synthetic samples through the fitting of a quadratic polynomial to the data points, which inherently captures more complex relationships between features. While SMOTE and ADASYN require the optimization of both the number of nearest neighbors (k) and the balancing ratio, quadratic interpolation requires the optimization of only the

imbalanced ratio. This simplification allows quadratic interpolation to concentrate on achieving the most effective balance between the minority and majority classes, ensuring that the synthetic data generated maintains the nuanced patterns present in the minority class without the additional complexity of tuning multiple parameters. This method aligns well with our goal of finding the optimum data balancing ratio in a cost-effective manner, providing high-quality synthetic data without the computational complexity of advanced methods. The application of quadratic interpolation underscores the innovative aspect of repurposing classical mathematical techniques for modern data balancing needs.

4 Optimization of data balancing ratio

Optimization is the process of adjusting or improving a system, process, or model to best achieve a specific goal. This goal is often related to minimizing a cost function or maximizing a gain function. In this study, the focus is on optimizing data balancing ratios in healthcare datasets to maximize classification accuracy while minimizing resource consumption. To date, no studies have addressed the optimization of data balancing ratios, despite its critical importance. Data balancing ratio optimization is crucial for improving model performance in addressing class imbalance problems. A balanced dataset allows classification algorithms to learn all classes accurately, enhancing performance metrics such as accuracy, precision, recall, and F1-score. Conversely, an improperly determined balancing ratio can result in poor performance on the minority class or overfitting due to excessive balancing. Finding an optimal ratio without overusing resources improves efficiency by reducing processing time, memory usage, and CPU consumption. Thus, balancing ratio optimization is a fundamental step for balanced learning and generalizability.

In this study, the results of three optimization methods—PSO, WOA, and Optuna—will be analyzed for data balancing ratio optimization. These meta-heuristic algorithms are widely used for optimization problems and are effective in identifying the optimal data balancing ratio. A key shared feature of these algorithms is their ability to solve nonlinear, multidimensional, and complex problems. Due to their stochastic nature, they explore a wide search space and are capable of escaping local minima to find global optimum solutions. Additionally, these algorithms offer resource-efficient solutions with low computational costs, making them advantageous for processes such as data balancing. Their flexible and easily adjustable parameters allow them to be readily applied to various datasets. In data balancing ratio optimization, these algorithms perform multi-objective optimization by considering not only classification accuracy but also resource consumption metrics such as processing time, memory usage, and CPU load. For example, PSO is advantageous for its rapid search capabilities, reducing processing time; Optuna excels in hyperparameter optimization, providing effective results; and WOA generates balanced solutions over a wide search space, enhancing performance while minimizing resource consumption. These features make these algorithms ideal choices for optimizing data balancing ratios.

4.1 PSO for data balancing optimization

PSO [71–73] is a computational method used for optimizing a wide range of functions by iteratively trying to improve candidate solutions with regard to a given measure of quality. PSO optimizes a problem by having a population (swarm) of candidate solutions (particles) move around in the search space according to simple mathematical rules based on the particle's position and velocity. Each particle's movement is influenced by its local best-known position and is also guided toward the best-known positions in the search space, which are updated as better positions are found by other particles.

In the context of addressing class imbalance in machine learning, PSO is particularly important as it provides an efficient and effective means of optimizing multiple parameters simultaneously. This is crucial for balancing datasets, where the choice of parameters such as the number of nearest neighbors (k) and the balancing ratio can significantly affect the performance of the classification model. Compared to traditional optimization methods, PSO offers several advantages, including ease of implementation, the ability to escape local optima, and the capability to handle large and complex search spaces. To prevent overfitting, several strategies were implemented. First, fivefold cross-validation was integrated into the evaluation process, ensuring the model's performance generalizes across different data subsets rather than overfitting to specific training data. A fivefold setup was determined through grid search, striking a balance between computational efficiency and robust evaluation. During the optimization, parameter combinations that performed well on the training data but poorly on the validation sets were penalized, ensuring that the final parameters yielded consistent results across all folds. This approach reduced the risk of overfitting and improved the model's generalizability. To avoid getting trapped in local minima, PSO employed strategies to maintain a balance between exploration and exploitation. The adaptive update mechanism of particle velocities allowed particles to explore diverse areas of the search space. For example, the inertia weight (w) was gradually reduced from 0.9 to 0.4 during optimization to shift from exploration to exploitation as the algorithm progressed. Additionally, randomness was introduced in velocity updates, and particles were periodically reset to random positions in the search space. This stochastic behavior increased diversity in the swarm and reduced the likelihood of premature convergence.

In the proposed study, a unique objective function was defined and implemented using PSO to optimize the parameters for various oversampling techniques, including SMOTE, ADASYN, SVM-SMOTE, Borderline-SMOTE, cubic interpolation, and quadratic interpolation. This objective function considers multiple performance metrics, including accuracy, processing time, memory usage, and CPU usage, to ensure a comprehensive evaluation and optimization of the data balancing process. The step-by-step application of PSO for optimization is like in the below;

1. *Define the Objective Function* The objective function for PSO is designed to optimize the parameters for the chosen oversampling technique. These parameters typically include the k value (number of nearest neighbors) for methods like SMOTE and ADASYN, and the balancing ratio for all methods. The function

evaluates the performance of these parameters by applying the oversampling technique to resample the data, training a SVM classifier, and then assessing the model based on multiple performance metrics.

2. *Initialize PSO Parameters* PSO requires the definition of several parameters, including the number of particles, the maximum number of iterations, and the bounds for each parameter being optimized. For instance, the lower bounds (lb) and upper bounds (ub) for the k value and the balancing ratio can be set like in Eq. 27.

$$lb = [1, 0.9], ub = [10, 1.1] \quad (27)$$

In the equation, the k value ranges from 1 to 10, and the balancing ratio ranges from 0.9 to 1.1.

3. *Evaluate and Normalize Performance Metrics* During each iteration, the objective function is evaluated for each particle. The performance metrics are then normalized to ensure they are on a comparable scale. The normalization formulas for memory usage, CPU usage, and processing time are given in Eqs. 28–30.

$$\text{normalized_memory_usage} = \frac{\text{memory_usage}}{\text{MAX_MEMORY_USAGE}} \quad (28)$$

$$\text{normalized_cpu_usage} = \frac{\text{cpu_usage}}{\text{MAX_CPU_USAGE}} \quad (29)$$

$$\text{normalized_time} = \frac{\text{elapsed_time}}{\text{MAX_TIME}} \quad (30)$$

In Eqs. 28–30, MAX_MEMORY_USAGE (1024MB) represents the maximum memory usage observed during the experiments, MAX_CPU_USAGE (100%) represents the maximum CPU usage observed during the experiments, and MAX_TIME (300 second) represents the maximum processing time observed during the experiments.

4. *Calculate Objective Function* In the calculation of the objective function for this study, a weighted sum of the normalized performance metrics is utilized, reflecting the relative importance of each factor in the overall model evaluation. The weights for each criterion—accuracy, processing time, memory usage, and CPU usage—are deliberately chosen based on the specific requirements and priorities of the study. The highest weight is assigned to accuracy ($w_1 = 0.7$), underscoring its paramount importance in classification tasks, especially in medical applications where the correct diagnosis can significantly impact patient outcomes. High accuracy ensures that the model can effectively differentiate between conditions, a critical factor in clinical settings. This heavy weighting reflects a focus on developing predictive models that not only perform well statistically but also deliver

practical, reliable results in real-world applications. The weights for processing time, memory usage, and CPU usage are set to 0.1 (w_2 , w_3 , w_4 , respectively), representing the computational efficiency of the model. These metrics, while secondary to accuracy, play a crucial role in ensuring that the model can operate efficiently, particularly in real-time or resource-constrained environments. The distribution of these weights acknowledges the necessity of balancing high predictive performance with the practical deployability of the model. The determination of these weights is guided by both domain-specific requirements and practical considerations. For example, the emphasis on accuracy is driven by the critical nature of medical diagnostics, where false positives or false negatives can lead to significant consequences. Conversely, the weights for computational efficiency metrics reflect the need for models that can function effectively on standard or resource-limited hardware often found in healthcare settings. These weights are designed to align the model's optimization priorities with real-world constraints, such as the processing capabilities of devices and the need for rapid, reliable predictions in clinical workflows. By allocating these weights, the study ensures a holistic optimization approach, emphasizing accuracy while addressing essential aspects of computational efficiency. This balance guarantees that the developed models are not only statistically robust but also practical for deployment, meeting the dual objectives of high predictive power and operational feasibility. The calculation of the objective function is given by Eq. 31.

$$\begin{aligned} \text{objective function} = \text{score} = & (w_1 \times \text{normalized_accuracy}) \\ & + (w_2 \times (1 - \text{normalized_time})) \\ & + (w_3 \times (1 - \text{normalized_memory_usage})) \\ & + (w_4 \times (1 - \text{normalized_cpu_usage})) \end{aligned} \quad (31)$$

Equation 31 has been uniquely designed to align with the objectives of this study, producing a fitness function that maximizes classification accuracy while also accounting for resource consumption. For instance, a software solution with extremely high accuracy but excessive memory and CPU usage would be undesirable, making it crucial to incorporate resource consumption into the fitness function, even to a minimal degree. In the equation, accuracy represents the classification accuracy obtained from the classifiers used in the study, while resource consumption indicates the amount of resources required by the method.

5. *Update Particle Positions and Velocities* The particles' positions and velocities are updated based on their own best-known positions and the global best-known position. This process is iterated, allowing the swarm to converge toward the optimal solution. In this study, we used a swarm size of 50 particles and ran the optimization for a maximum of 100 iterations. The parameters for the PSO algorithm, including the swarm size and the number of iterations, were determined through a grid search process to identify the most effective configuration for our specific dataset and objective function. During the grid search, the swarm size was evaluated over the range [10, 20, 30, 40, 50] and the maximum number of itera-

tions was tested within [50,100,150,200]. These ranges were chosen to balance computational efficiency and optimization effectiveness. Smaller swarm sizes and fewer iterations were considered to minimize computational costs, while larger values were included to ensure thorough exploration of the search space and convergence to the global optimum. The chosen configuration of a swarm size of 50 and 100 iterations provided the best trade-off between computational efficiency and the quality of the solutions. This setup allowed for sufficient exploration of the parameter space while maintaining reasonable execution times, ensuring that the PSO algorithm effectively optimized the objective function under the constraints of the study. Also, the parameters for the PSO algorithm, including the swarm size, number of iterations, inertia weight (w), personal learning coefficient ($c1$), and social learning coefficient ($c2$), were determined through a grid search process. Specifically, the inertia weight (w) was searched within [0.4,0.9], and the personal ($c1$) and social learning coefficients ($c2$) were evaluated within [1.5, 2.5]. These ranges were chosen to balance exploration and exploitation, ensuring the algorithm could escape local minima while efficiently converging to the global optimum. After the grid search, the optimal values were determined to be $w=0.7$, $c1=2$, and $c2=2$. This configuration provided the best trade-off between computational efficiency and the quality of the solutions, ensuring effective optimization of the dataset balancing ratio while minimizing resource consumption.

6. *Select Optimal Parameters* The PSO algorithm identifies the optimal parameters for each oversampling technique that maximize the overall score. These optimal parameters are those that yield the highest score based on the defined objective function.

PSO is chosen for its efficiency in optimizing multiple parameters simultaneously. While the PSO algorithm itself is not new, our study introduces a novel fitness function that incorporates accuracy, processing time, memory usage, and CPU usage. By using PSO, the proposed study effectively optimizes the parameters for various oversampling methods, ensuring that each balancing method is tuned for the best possible performance across multiple metrics. This comprehensive approach ensures that the data balancing ratio is optimized not only for performance but also for cost-effectiveness and computational efficiency. The novelty lies in the application of PSO to find the optimum balancing ratio tailored to the unique characteristics of each dataset. This approach directly aligns with our goal of achieving practical and efficient data balancing solutions.

4.2 WOA for optimization of data balancing ratios

WOA [74–76] is a computational meta-heuristic method inspired by the social hunting behavior of humpback whales. WOA optimizes a wide range of functions by simulating bubble-net feeding strategies, where candidate solutions (whales) iteratively update their positions based on the best-known solution and their relative positions within the search space. Each whale's movement is influenced by exploration

(searching for new solutions globally) and exploitation (refining the best-known solutions locally), making WOA particularly effective for navigating complex search spaces and finding global optima.

In the context of addressing class imbalance in machine learning, WOA is highly significant as it provides an efficient mechanism for optimizing multiple parameters simultaneously. This is particularly crucial for balancing datasets, where parameters such as the number of nearest neighbors (k) and the balancing ratio can significantly influence the classification model's performance. Compared to traditional optimization methods, WOA offers several advantages, including simplicity of implementation, robust exploration capabilities, and the ability to avoid premature convergence to local optima. To avoid local optima, WOA employs a dynamic balance between exploration and exploitation. In the exploration phase, whales move randomly in the search space to explore new areas, ensuring diverse solutions. This was further enhanced by introducing adaptive parameters like a (which decreases linearly from 2 to 0 during iterations) to adjust the exploration and exploitation behavior dynamically. Additionally, the initial positions of whales were randomized across the entire search space, increasing the probability of escaping local optima. For example, in this study, the range for k value ([1, 10]) and the balancing ratio ([0.9, 1.1]) was deliberately chosen to cover a broad space for parameter optimization. To prevent overfitting, a fivefold cross-validation strategy was integrated into the evaluation process. By splitting the data into five folds, the model was trained and validated on different subsets of the data, ensuring that the performance metrics reflected the model's generalization capabilities rather than its performance on a specific training set. This approach significantly reduced the risk of overfitting to specific data configurations. For instance, during optimization, parameter configurations that showed high accuracy but poor generalization across folds were penalized, ensuring that only robust solutions were selected.

In this study, WOA is applied with a unique objective function to optimize the parameters for various oversampling techniques, including SMOTE, ADASYN, SVM-SMOTE, Borderline-SMOTE, cubic interpolation, and quadratic interpolation. The objective function incorporates multiple performance metrics—accuracy, processing time, memory usage, and CPU usage—to ensure a comprehensive evaluation and optimization of the data balancing process. These metrics are normalized to make them comparable, and weights are assigned based on their relative importance. Accuracy ($w_1 = 0.7$) is given the highest weight due to its critical role in medical applications, where accurate classification can have significant real-world implications. Computational efficiency metrics—processing time, memory usage, and CPU usage—are each assigned weights of 0.1 (w_2, w_3, w_4 , respectively), reflecting the need for practical and deployable solutions.

The optimization process involves the following steps:

1. *Define the Objective Function* Objektif fonksiyon PSO ile aynı yapıdadır ve bölüm 9.1'de verilmektedir.
2. *Normalize Performance Metrics* During each iteration, the objective function is evaluated for each particle. The performance metrics are then normalized to ensure they are on a comparable scale. The normalization formulas for memory usage, CPU usage, and processing time are given in Eqs. 28–30.

3. *Calculate the Objective Function* A weighted sum of normalized metrics is used to compute the objective score using Eq. 31.
4. *Update Whale Positions* Whales update their positions based on their proximity to the current best solution. Randomly selected positions and adaptive exploration strategies help diversify the search and avoid premature convergence. Position updates are provided using Eqs. 32–34.

$$X(t+1) = X^*(t) - A \cdot |C \cdot X^*(t) - X(t)| \quad (32)$$

$$A = 2 \cdot a \cdot r - a \quad (33)$$

$$C = 2 \cdot r \quad (34)$$

In the equations, $X(t+1)$ represents the whale's next position, $X^*(t)$ denotes the best position in the swarm, and $X(t)$ is the whale's current position. The parameters A and C are adaptive control parameters, a is a parameter that decreases throughout the algorithm, and r is a random number within the range $[0, 1]$.

5. *Select Optimal Parameters* During the optimization, the parameters of the WOA algorithm, such as the population size, the number of iterations, and the control parameter a (which governs the exploration and exploitation phases), are fine-tuned through grid search. The population size is tested within $[10, 20, 30, 40, 50]$, and the maximum number of iterations is explored in the range $[50, 100, 150, 200]$. Additionally, the control parameter a is evaluated within the range $[0.5, 2.0]$ to balance the exploration and exploitation phases effectively. The optimal configuration—50 whales, 100 iterations, and $a = 1.5$ —was found to provide the best trade-off between computational efficiency and optimization quality, ensuring that WOA effectively identified the ideal balancing ratio under the constraints of the study.

WOA is chosen for its robust performance in global optimization tasks, particularly in balancing exploration and exploitation. This study's novelty lies in defining a unique fitness function that integrates accuracy and computational efficiency metrics, tailoring the optimization process to the specific requirements of medical datasets. By leveraging WOA, the study achieves not only optimal data balancing ratios but also cost-effective and computationally efficient solutions, directly aligning with the goal of practical and deployable data balancing techniques.

4.3 Optuna for optimization of data balancing ratios

Optuna [77, 78] is a modern and flexible hyperparameter optimization framework that adaptively explores the search space using advanced techniques like Bayesian optimization and tree-structured Parzen estimators (TPE). Its ability to dynamically refine parameter searches and focus on promising regions makes it highly effective for complex tasks like data balancing optimization. In this study, Optuna was used to tune key parameters for oversampling techniques, including SMOTE, ADASYN,

SVM-SMOTE, Borderline-SMOTE, cubic interpolation, and quadratic interpolation, ensuring optimal performance across multiple metrics such as accuracy, processing time, memory usage, and CPU usage.

Optuna integrates several mechanisms to mitigate overfitting during the optimization process. Firstly, k-fold cross-validation was employed to evaluate each hyperparameter combination, ensuring the model's performance generalizes across different data subsets. A fivefold cross-validation strategy was chosen through grid search to balance computational cost and robust evaluation, reducing the risk of overfitting to a specific training set. Additionally, Optuna's pruner mechanisms, such as the Median Pruner and Successive Halving, were utilized to halt underperforming trials early. This prevents wasting computational resources on suboptimal solutions and focuses the optimization process on promising configurations. Furthermore, the objective function was designed to penalize overly complex solutions or excessive oversampling ratios, ensuring that the resulting parameters lead to balanced datasets without introducing noise or bias.

Optuna's dynamic approach to search space exploration helps avoid local minima. By leveraging Bayesian optimization with TPE, Optuna adaptively explores the parameter space, focusing on high-potential regions based on prior evaluations. This adaptive strategy ensures a thorough search of the global solution space while refining promising areas. To further encourage exploration, Optuna incorporates random sampling in early trials, introducing diversity to the optimization process. This stochastic component prevents the algorithm from converging prematurely to suboptimal solutions. Additionally, Optuna's ability to dynamically adjust the search space based on observed performance allows for targeted exploration of promising regions while remaining flexible to explore new areas.

Optuna's adaptive and dynamic optimization process effectively balances exploration and exploitation, ensuring robust parameter tuning for oversampling techniques. Its integration of cross-validation, random sampling, and pruning mechanisms minimizes overfitting and avoids local minima. By leveraging these capabilities, the study achieved practical and efficient data balancing solutions tailored to the specific requirements of each dataset, aligning with the overall goal of developing deployable and high-performing machine learning models.

5 Classification of datasets

In fact, the classification process holds a lower priority in our method. This is because, regardless of the classifier used, it is possible to measure the contribution of data balancing ratios to classification performance. In this study, SVM, EL, and RF classifiers were used across three different balancing scenarios (full balance, optimized balance, and imbalance). These classifiers were tested individually to comprehensively analyze the contribution of data balancing optimization to their performance. These classifiers were chosen to evaluate their capacity to learn and generalize class distinctions in the dataset due to their diverse structures and algorithmic approaches. SVM is a robust method for learning nonlinear separations, providing high accuracy by maximizing the decision boundaries between classes. EL improves generalization performance and produces noise-resistant results by

combining multiple models to enhance diversity. RF combines multiple decision trees, reducing error rates and offering a mechanism to prevent overfitting. These classifiers were assessed for their potential superiority in handling different data structures, allowing the study to provide multiple perspectives on model selection and enabling a comprehensive analysis. This approach is crucial for understanding which classifier performs best under varying conditions.

5.1 SVM for classification

SVM [79–81] is a powerful and versatile machine learning method used for classification and regression tasks. SVMs work by finding the optimal hyperplane that separates the classes in the feature space, maximizing the margin between the data points of different classes. This characteristic allows SVMs to be particularly effective in high-dimensional spaces and in scenarios where the number of dimensions exceeds the number of samples. SVMs are known for their robustness, ability to handle nonlinear relationships through the use of kernel functions, and their efficiency in both small and large datasets. Due to these advantages, SVMs are widely used in various fields, including bioinformatics, text classification, and image recognition, making them a critical tool for building predictive models in machine learning.

In this study, while SVM serves as the classifier, the primary focus is on optimizing data balancing methods. The importance of this research lies in demonstrating how different data balancing techniques, when optimized, can significantly impact the performance of classification models. The study goes beyond evaluating the performance of the SVM classifier alone, emphasizing the crucial role of data balancing in handling class imbalance. By optimizing parameters such as the k value and the balancing ratio using optimization methods (PSO, WOA, Optuna), this study aims to reveal how fine-tuning these methods can enhance the overall accuracy, efficiency, and robustness of predictive models. The results underscore the significance of selecting and optimizing data balancing techniques to improve the reliability and fairness of machine learning applications, especially in domains with imbalanced datasets. The SVM classifier is applied in this study to evaluate the effectiveness of various oversampling techniques. Here is a detailed step-by-step guide on how SVM is utilized:

1. *Data Preparation* The each dataset used in this study is split into training and test sets for each dataset individually. The dataset splitting process is conducted using fivefold cross-validation, which ensures that the model is trained and validated on multiple subsets of the data, thereby providing a robust evaluation of its performance. The choice of five folds was determined through a grid search process, balancing computational efficiency with reliable performance estimation. Cross-validation not only ensures that the model is trained on representative portions of the data but also reduces the risk of overfitting by exposing the model to different subsets during training and validation. This systematic approach ensures that the final evaluation metrics reflect the model's generalization capability, making it well suited for real-world applications.

2. *Feature Scaling* Feature scaling is performed to normalize the data. This step is crucial for SVM performance because SVMs are sensitive to the scale of input features. Normalization ensures that all features contribute equally to the model's decision boundary. The scaling is typically done using Eq. 35.

$$X_{scaled} = \frac{X - \mu}{\sigma} \quad (35)$$

In Eq. 35, X is the original feature vector, μ is the mean of the training data, and σ is the standard deviation of the training data.

3. SVM classifier with a radial basis function (RBF) kernel is trained using the training data. Hyperparameter tuning is performed to find the best combination of parameters C (regularization parameter) and γ (kernel coefficient). The objective is to find the optimal values that minimize the cost function in Eq. 36.

$$\text{cost} = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i; C, \gamma)) \quad (36)$$

In Eq. 36, n represents the number of samples in the validation set, indicating how many individual errors are summed to compute the overall cost. L is the loss function, specifically chosen based on the requirements of the SVM and its formulation, such as hinge loss commonly used with SVMs. y_i are the actual labels of the data points, used in conjunction with predictions to determine the loss. $f(x_i; C, \gamma)$ denotes the SVM model's prediction for the input x_i parameterized by C and γ , reflecting how the model uses these parameters to calculate the decision boundary. This cost function aims to optimize the parameters C and γ to minimize the total validation loss, thereby enhancing the model's ability to generalize effectively to new, unseen data. The process of hyperparameter tuning typically employs strategies like grid search or random search to explore various combinations of C and γ , seeking the set that achieves the lowest possible cost on the validation data.

4. *Model Evaluation* The best model identified by the hyperparameter tuning process is evaluated on the test set. The SVM classifier with a radial basis function (RBF) kernel is trained using the training data. Hyperparameter tuning is performed to find the best combination of parameters: C (regularization parameter) and γ (kernel coefficient). These parameters were optimized through a grid search process, ensuring the model achieves optimal performance. The search ranges for C and γ were carefully selected to balance model complexity and generalization. Specifically, C was tested in the range [0.1, 1, 10, 100, 1000] and selected as 0.1, and γ was evaluated within [0.001, 0.01, 0.1, 1, 10] and selected as 0.1. This systematic approach allowed the identification of the best parameter combination by evaluating performance across multiple configurations. By leveraging grid search, the SVM classifier was fine-tuned to effectively model complex patterns in the data, ensuring both high accuracy and robustness.

SVM is selected as the classifier due to its robustness and effectiveness in high-dimensional spaces. By following these steps, the SVM classifier is applied systematically to evaluate and optimize the performance of various oversampling techniques. This process ensures that the data balancing methods are tuned effectively, leading to more accurate and robust classification models. The choice of SVM highlights the study's focus on demonstrating that even classical classifiers can achieve high accuracy when combined with optimized data balancing techniques. This approach underscores the feasibility of achieving effective results with more accessible and cost-effective methods, further validating the proposed data balancing strategies.

5.2 RF for classification

RF [82–84] is a robust and versatile machine learning method used for both classification and regression tasks. RF operates by building an ensemble of decision trees, where each tree contributes to the final prediction by voting or averaging, depending on the task. This ensemble approach makes RF particularly effective at reducing overfitting, enhancing its ability to generalize well on unseen data. RF is also adept at handling high-dimensional spaces and datasets with missing values, making it a widely used method across various fields, including health care, bioinformatics, and finance. Its built-in feature importance metric allows for the identification of the most relevant predictors, further increasing its utility in practical applications.

In this study, while RF serves as the classifier, the primary focus is on optimizing data balancing methods. The importance of this research lies in demonstrating how different data balancing techniques, when optimized, can significantly impact the performance of classification models. The study goes beyond evaluating the performance of the RF classifier alone, emphasizing the crucial role of data balancing in handling class imbalance. By optimizing parameters such as the number of trees, maximum depth, and balancing ratio using optimization methods (PSO, WOA, Optuna), this study aims to reveal how fine-tuning these methods can enhance the overall accuracy, efficiency, and robustness of predictive models. The results underscore the significance of selecting and optimizing data balancing techniques to improve the reliability and fairness of machine learning applications, especially in domains with imbalanced datasets. The RF classifier is applied in this study to evaluate the effectiveness of various oversampling techniques. Here is a detailed step-by-step guide on how RF is utilized:

1. *Data Preparation* Each dataset used in this study is split into training and test sets. The dataset splitting process is conducted using fivefold cross-validation, ensuring that the model is trained and validated on multiple subsets of the data. This systematic approach reduces the risk of overfitting and ensures robust evaluation metrics. The choice of five folds was determined through a grid search process, balancing computational efficiency with reliable performance estimation. Cross-validation ensures that the model is trained on representative portions of the data, providing a reliable estimate of its generalization capability.

2. *Feature Scaling* While RF is not inherently sensitive to the scale of features, standardization is performed in this study to ensure uniformity across preprocessing steps, particularly since other classifiers involved in the comparison require feature scaling. The scaling is typically done using Eq. 37.

$$X_{scaled} = \frac{X - \mu}{\sigma} \quad (37)$$

In Eq. 37, X represents the original feature vector, μ is the mean of the training data, and σ is the standard deviation of the training data.

3. *Training and Hyperparameter Tuning* The RF classifier is trained using the training data. Hyperparameter tuning is performed to find the optimal combination of parameters, such as the number of trees ($n_{estimators}$), maximum depth (d_{max}), and minimum samples per split (s_{min}). The objective is to minimize the cost function defined using Eq. 38.

$$\text{cost} = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i, s_{min}, d_{max})) \quad (38)$$

In Eq. 38, n represents the number of samples in the validation set, L is the loss function, such as log loss or Gini impurity, depending on the RF formulation, y_i represents the actual labels of the data points, $f(x_i, s_{min}, d_{max})$ denotes the RF model's prediction for input x_i based on the selected parameters.

The hyperparameter tuning process explores a range of values for $n_{estimators}$, d_{max} , and s_{min} using grid search or random search. $n_{estimators}$ was tested in the range [10, 50, 100, 200], with the best value determined to be 100. d_{max} was evaluated within [5, 10, 20, none], and the best result was found with 10. s_{min} was optimized within [2, 5, 10], with 5 providing the best performance. This systematic tuning ensures that the RF classifier is both accurate and efficient.

4. *Model Evaluation* The best RF model identified through hyperparameter tuning is evaluated on the test set. Metrics, such as accuracy, precision, recall, specificity, and F1-score, are calculated to assess the classifier's performance comprehensively. By leveraging grid search and robust evaluation metrics, the RF classifier is fine-tuned to handle complex patterns in the data while addressing class imbalance through optimized balancing RF is selected as the classifier due to its robustness, ability to handle imbalanced datasets, and effectiveness in high-dimensional spaces.

By following these steps, the RF classifier is applied systematically to evaluate and optimize the performance of various oversampling techniques. This process ensures that the data balancing methods are tuned effectively, leading to more accurate and robust classification models. The choice of RF highlights the study's focus on demonstrating that even classical classifiers can achieve high accuracy when

combined with optimized data balancing techniques. This approach underscores the feasibility of achieving effective results with more accessible and cost-effective methods, further validating the proposed data balancing strategies.

5.3 Ensemble learning for classification

EL [85–87] are a powerful category of machine learning algorithms that combine predictions from multiple base models to enhance overall performance. These methods, including bagging, boosting, and stacking, reduce variance, bias, and improve the stability of predictions. Ensemble techniques are particularly effective in addressing challenges posed by imbalanced datasets and high-dimensional feature spaces, making them invaluable tools in fields such as health care, bioinformatics, and finance. Notable examples include RF and gradient boosting machines (GBM), which are widely adopted for their robustness and flexibility. In this study, while ensemble methods serve as the classifiers, the primary focus is on optimizing data balancing methods. The importance of this research lies in demonstrating how different data balancing techniques, when optimized, can significantly impact the performance of classification models. The study goes beyond evaluating the performance of ensemble classifiers alone, emphasizing the crucial role of data balancing in addressing class imbalance. By optimizing parameters such as the number of estimators, learning rates, and balancing ratios using optimization methods (PSO, WOA, Optuna), this study aims to reveal how fine-tuning these methods can enhance the overall accuracy, efficiency, and robustness of predictive models. The results underscore the significance of selecting and optimizing data balancing techniques to improve the reliability and fairness of machine learning applications, especially in domains with imbalanced datasets. Below is a detailed step-by-step guide on how ensemble methods are utilized in this study:

1. *Data Preparation* Each dataset used in this study is split into training and test sets. The dataset splitting process is conducted using fivefold cross-validation to ensure robust evaluation. Cross-validation exposes the model to different subsets of data, reducing overfitting and providing reliable estimates of the model's generalization capability. The choice of five folds was determined through a grid search process, balancing computational efficiency with robust performance evaluation. This systematic approach ensures that the results are reliable and suitable for real-world applications.
2. Although ensemble methods are generally insensitive to feature scaling, normalization is performed in this study to ensure consistency across preprocessing steps, particularly since other classifiers in the study require normalized data. Feature scaling is performed using Eq. 37.
3. *Model Training and hyperparameter tuning* Ensemble methods, such as RF, gradient boosting, or stacking, are trained using the training data. Hyperparameter tuning is performed to find the optimal combination of parameters like the number of estimators ($n_{estimators}$), learning rate (α), and maximum depth (d_{max}). The

objective is to minimize a cost function that evaluates the model's error on the validation set, defined using Eq. 39.

$$\text{cost} = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i, n_{\text{estimators}}, \alpha, d_{\text{max}})) \quad (39)$$

In Eq. 39, n represents the number of samples in the validation set, L denotes the loss function such as log loss or squared error, y_i corresponds to the actual labels of the data points, and $f(x_i, n_{\text{estimators}}, \alpha, d_{\text{max}})$ is the ensemble model prediction. Grid search or random search is employed to systematically optimize the hyperparameters. For instance, $n_{\text{estimators}}$ is tested in the range [10, 50, 100, 200], with the optimal value identified as 100. Learning rates (α) are evaluated within [0.001, 0.01, 0.1, 1], with 0.1 selected as the optimal value. The maximum depth (d_{max}) is optimized within [5, 10, 20, none], and 10 provides the best performance. This systematic hyperparameter tuning ensures that the ensemble classifier achieves accuracy, efficiency, and robustness, enabling an effective evaluation of the proposed data balancing techniques.

Ensemble methods are a natural choice in this study due to their ability to leverage the strengths of multiple base models, providing robust and reliable predictions. Unlike single-model approaches such as RF, which relies on aggregating the predictions of individual decision trees built on bootstrap samples, ensemble methods like gradient boosting or stacking combine diverse models in a more sequential or hierarchical manner. Gradient boosting, for example, builds models iteratively, focusing on correcting the errors of previous models, while stacking leverages multiple classifiers or regressors and combines their outputs through a meta-model, often resulting in improved predictive accuracy. These advanced ensemble techniques excel in capturing complex patterns in data and are particularly effective in addressing challenging imbalanced datasets. The research demonstrates that even classical ensemble methods, when combined with optimized data balancing techniques, can achieve exceptional results. By systematically fine-tuning hyperparameters and balancing strategies, this approach highlights the feasibility of achieving high performance without relying on overly complex or resource-intensive methods. These findings validate the proposed balancing strategies and underscore their potential for broader applications in machine learning and data science, particularly by offering flexibility and adaptability beyond RF's traditional bagging-based approach.

6 Performance metrics

The performance of a method is not solely determined by its classification accuracy. A review of the literature reveals that most studies focus primarily on accuracy, often neglecting runtime and resource management in their proposed methods. This study specifically examines the impact of data balance ratio optimization on classifiers. For the first time in the literature, this discussion includes not only metrics that measure quality—such as accuracy, precision, recall, and F1-score—but also

metrics that assess resource consumption, including runtime, CPU usage, and memory usage. The primary reason for calculating these metrics is to evaluate whether the proposed approach is practical for real-world applications. In healthcare settings, professionals often run multiple programs simultaneously, and any additional method implemented in these systems would impose an additional load. Therefore, methods that require excessive time or resources are unlikely to be adopted. For these reasons, the performance evaluation in this study offers researchers a new perspective, emphasizing the importance of resource consumption in future studies. This focus encourages the consideration of resource efficiency alongside classification accuracy. This section provides detailed explanations of the performance metrics used in this study.

1. *Accuracy* Accuracy [88] measures the overall effectiveness of the model by quantifying the proportion of true results (both true positives and true negatives) among the total number of cases examined. This metric is paramount, especially in medical applications, as it provides a general indication of how well the model performs across all classifications, highlighting its reliability in diagnosing conditions correctly. The accuracy metric is calculated using Eq. 40.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (40)$$

This metric provides a general measure of the model's performance but may not fully capture the performance on imbalanced datasets, where other metrics become crucial.

2. *Precision* Precision [88] indicates the correctness of positive predictions, measuring the proportion of actual positives among the positive calls made by the model. This metric is critical in medical diagnostics where the cost of false positives (such as incorrect disease diagnosis) can lead to unnecessary treatments or anxiety for patients. High precision in a model ensures that the diagnoses made are accurate, minimizing the risk of false alarms. The precision metric is calculated using Eq. 41.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (41)$$

In the equation, "True Positives" refers to the instances where the model correctly predicts the positive class, while "False Positives" are instances where the model incorrectly predicts the positive class for actual negatives.

3. *Recall*(Sensitivity or True Positive Rate): Recall [88] (also known as sensitivity) assesses the model's ability to identify all actual positive cases. In the context of disease diagnosis, a high recall rate is crucial as it reduces the risk of missing a true positive case, such as failing to identify a disease condition, which can be detrimental to patient health. Recall is especially important when the consequences

of missing a positive diagnosis are severe. The recall metric is calculated using Eq. 42.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (42)$$

In Eq. 42, "False Negatives" refer to the instances where the model incorrectly predicts the negative class for actual positives, meaning it fails to detect conditions that are actually present.

4. *F1-Score* F1-score [88] provides a balance between precision and recall, offering a single measure that harmonizes the two metrics. It is particularly useful when seeking a balance between the necessity of not missing positives (high recall) and ensuring that the positives identified are indeed correct (high precision). The F1-score is beneficial in scenarios where both false positives and false negatives carry significant consequences. The F1-score metric is calculated using Eq. 43.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (43)$$

The F1-score is particularly useful for evaluating the performance of models on imbalanced datasets, where both false positives and false negatives are important to consider.

5. *Processing Time* Processing time, as utilized in this study, specifically measures the time required to balance the dataset using the proposed data resampling methods. This metric is crucial for assessing the efficiency of the data balancing technique, highlighting how quickly data can be prepared for model training. This aspect is particularly important in real-time or resource-constrained environments where the speed of data preparation directly impacts the overall time to decision-making. By focusing on the duration needed for data resampling, the study aims to evaluate the practicality of the data balancing methods, ensuring they are not only effective in improving model performance but also feasible in terms of their operational demands. This focus helps in determining whether the proposed methods can be seamlessly integrated into workflows where quick turnaround is essential. The processing time is calculated using Eq. 44.

$$\text{elapsed_time}_{\text{data_balance}} = \text{end_time}_{\text{data_balance}} - \text{start_time}_{\text{data_balance}} \quad (44)$$

6. *Memory Usage* Memory usage is monitored during the execution of the method to ensure that the approach is resource-efficient. High memory usage can be a bottleneck in large-scale or real-time applications, making this metric crucial for practical deployment. The memory usage as MegaBytes (MB) is calculated using psutil function in Python.
7. *CPU Usage* CPU usage is tracked to evaluate the computational load imposed by the method. This metric helps in understanding the scalability and feasibility of the proposed method in different computational settings. It indicates the percent-

age of CPU resources consumed during the model training and evaluation phases. The CPU usage is calculated as percentage using `psutil` function in Python.

7 Evaluation results

Data balancing is the process of addressing class imbalances in datasets by generating new samples for the minority class to mitigate disparities between classes. When a dataset has an imbalanced distribution, classifiers tend to learn the majority class better while failing to adequately learn the minority class. This leads to poor classification performance during testing, resulting in inaccurate separation of classes and an inconsistent model. A review of the literature reveals that data balancing methods are often limited to specific approaches and typically aim to equalize the number of samples across classes. In this study, two novel contributions are made: first, the adaptation of interpolation-based methods, cubic and quadratic interpolation, to the data balancing problem for the first time; and second, focusing on the optimization of class distribution to maximize classification performance rather than strictly equalizing class sizes.

In the healthcare domain, datasets often exhibit significant class imbalances between the number of samples from patients and healthy individuals. Prioritizing data balancing and assessing its impact on classification performance in studies in this field is crucial, as an individual's health is invaluable. In this study, data balancing was performed on six different datasets (Kidney, Z-Alizadeh Sani, WDBC, WPBC, Stroke, and Diabetes) using both interpolation-based methods and widely used techniques in the literature, such as SMOTE, ADASYN, Borderline-SMOTE, and SVM-SMOTE. The datasets were selected from the limited number of available open datasets, prioritizing those with high imbalance ratios. The primary aim of this study is to optimize the balancing ratio to maximize classifier performance. For the optimization process, three different methods—particle swarm optimization (PSO), whale optimization algorithm (WOA), and Optuna—were tested, and the response of classifier performance to these optimization methods was evaluated. A key aspect of the optimization process is that the fitness function considers accuracy, time, CPU usage, and memory usage together. Methods that achieve high classification accuracy but consume excessive resources are not practical in real-world applications. Therefore, this study addresses an issue often overlooked in the literature. Performance evaluations were conducted under three different scenarios: when the class distribution was imbalanced, when it was fully balanced, and when the optimized balance ratio was achieved. These scenarios were tested for each dataset using three classifiers—support vector machine (SVM), ensemble learning, and random forest. Performance metrics, including accuracy, precision, recall, F1-score, time, CPU usage, and memory usage, were used to evaluate the results. This evaluation is the first of its kind to assess both performance and resource consumption, offering valuable insights into the practicality of the methods. The evaluations for each dataset are presented under separate subheadings in this section.

7.1 Performance evaluation for WDBC dataset (SET1)

Breast cancer is a common type of cancer among women and can be treated effectively with early diagnosis. While breast images can be used for diagnosis, clinical results, which enable faster diagnosis, are also utilized. The WDBC dataset was introduced as an open dataset to develop models for breast cancer diagnosis. Details about this dataset are provided in Sect. 2.1.1. The WDBC dataset was chosen for this study due to its imbalanced distribution of samples and the necessity for approaches to improve diagnostic systems for such a widespread condition. The WDBC dataset has an imbalance ratio of approximately 0.31, which is a critical factor that can affect classification performance. In this study, the WDBC dataset will be referred to as Set1. For Set1, the details of the best-performing combinations for each balancing method, optimization method, and classifier will be provided. Performance evaluation metrics will include the balancing ratio, accuracy, time, CPU usage, memory usage, and the fitness score. The fitness score, calculated using Eq. 2, indicates which method has the most significant impact on classification performance. The results of these calculations are presented in Table 1, where the top three outcomes are highlighted. The best result is marked in bold, the second-best in bolditalic, and the third-best in italic. Additionally, all optimization methods were tested for each classifier, but only the optimization method yielding the best balancing result is listed under the best optimizer heading.

Table 1 provides the results for each combination applied to Set1, the WDBC dataset. It shows that the optimized balance ratio for WDBC was achieved using the WOA optimization method, resulting in a balance ratio of 0.93, as calculated using Eq. 2. In the original dataset, the minority class contained 212 samples, while the majority class had 357 samples. At a balance ratio of 0.93, the number of samples in the minority class increased to 332, while the majority class remained unchanged. In data balancing, the quality of generated samples is as critical as their quantity, as quality determines how well the generated samples integrate with the minority class. Table 1 reveals that the quadratic interpolation method, adapted to this field for the first time, combined with WOA optimization and classified using ensemble learning, produced the best results for both accuracy and resource consumption. The fitness score, calculated to identify the most effective method, also highlights the potential of cubic interpolation as a valuable balancing technique. Moreover, the results indicate that balance ratio optimization does not burden the system and can operate efficiently on simple systems. In health care, every individual is highly significant, and even minor improvements can save lives, underscoring the necessity for such advancements. Table 1 demonstrates the critical importance of evaluating both accuracy and resource consumption in balance ratio optimization processes.

7.2 Performance evaluation for Z-Alizadeh Sani dataset (SET2)

Coronary artery diseases (CAD) represent one of the most significant risks to human life, with the number of patients succumbing to this condition increasing steadily

Table 1 Evaluation results for WDBC dataset

Dataset	Data balance method	Classification method	Best optimizer	Balancing ratio	Accuracy	Time (second)	Memory (MB)	CPU (%)	Fitness score
SET1	Quadratic	Ensemble learning	WOA	0.93	1	1.67	9.35	25	0.554
		Random forest	Optuna	1.05	0.9795	2.55	9.46	25.3	0.5298
	Cubic	SVM	Optuna	1.03	0.9931	6.08	9.06	24.2	0.5444
		Ensemble learning	WOA	0.93	1	3.27	9.42	25.2	0.5428
Borderline-SMOTE	Random forest	Random forest	WOA	0.93	0.9928	3.76	9.42	25.2	0.5355
		SVM	PSO	1.02	0.9862	4.01	9.06	24.2	0.5491
	Ensemble learning	Ensemble learning	Optuna	1	0.9813	0.29	9.57	25.6	0.5356
		Random forest	PSO	0.97	0.9903	2.54	10.2	27.2	0.4999
SMOTE	SVM	SVM	Optuna	1	0.9767	1.71	8.34	22.3	0.5203
		Ensemble learning	PSO	0.9859	1	0.84	9.67	25.9	0.5405
	Random forest	Random forest	Optuna	1	0.9813	1.23	9.35	25.6	0.5333
		SVM	WOA	0.9467	0.9905	2.3	9.67	25.9	0.5271
SVM-SMOTE	Ensemble learning	Ensemble learning	WOA	0.91	1	0.65	9.53	25.5	0.5491
		Random forest	WOA	0.93	1	1.26	9.46	25.3	0.5501
	SVM	SVM	PSO	0.95	1	0.3	9.65	25.8	0.5447
		Ensemble learning	WOA	0.89	0.95	1.1	10	26	0.52
ADASYN	Random forest	Random forest	WOA	0.91	0.955	1.5	9.8	25.5	0.53
		SVM	PSO	0.94	0.96	0.5	10.2	26.5	0.54

due to various factors. Therefore, early diagnosis and improvements that can assist in the detection of such diseases are urgently needed. For these reasons, the Z-Alizadeh Sani dataset has been selected as another dataset in this study. This dataset is rich in clinical data related to CAD patients and exhibits a notable imbalance, with a class imbalance ratio of 0.4. In this study, the Z-Alizadeh Sani dataset is referred to as Set2. The results for Set2 are presented in Table 2, where the best result is highlighted in bold, the second-best in bolditalic, and the third-best in italic.

The primary objective of our study is to achieve improvements in disease diagnosis while testing different combinations. In the healthcare domain, data collection often neglects the balance of class distributions, a factor that adversely affects classification performance. Table 2 presents the results of various combinations tested to address this issue. For Set2, the balance ratio that optimizes both resource consumption and accuracy is found to be 1.04. Moreover, the results indicate that the best outcomes are achieved using quadratic interpolation as the balancing method, ensemble learning as the classifier, and WOA for balance ratio optimization. Similarly, when examining the second- and third-best results, cubic interpolation emerges as a noteworthy balancing method. These findings suggest that, unlike linear sample generation methods (e.g., Borderline-SMOTE, SVM-SMOTE, SMOTE, and ADASYN), the nonlinear methods adapted in this study (cubic and quadratic interpolation) contribute more significantly to diagnostic improvements in health care. The primary reason for this is the nonlinear distribution of the minority class in Set2, which makes nonlinear generation methods more effective in enhancing performance.

7.3 Performance evaluation for diabetes (SET3)

Diabetes is a disease that adversely affects people's lives regardless of age. If not diagnosed early, it can negatively impact all organs in the human body, putting lives at risk. Similar to the other datasets in this study, this dataset was chosen because diabetes is a significant disease affecting human health and because it exhibits a level of imbalance where the effects of balancing methods can be effectively demonstrated. The dataset has an imbalance ratio of 0.53, making any improvement in this area crucial for human life. Detailed information about the dataset is provided in Sect. 2.1.3. In line with the primary objective of this study, the optimization of different combinations has been applied to this dataset as well. The results for the dataset, referred to as Set3, are presented in Table 3.

Table 3 contains performance evaluations for each combination. The table highlights the best result in bold, the second-best in bolditalic, and the third-best in italic. Upon examining the results, it is evident that the combination of Borderline-SMOTE and the SVM classifier with a balancing ratio of 1.04 yields the best performance. This result is achieved with very low resource consumption, including 0.48 s for runtime, 2.8% CPU usage, and 2.8 MB memory usage, indicating that the optimization process does not negatively impact resource consumption. The fitness scores presented in the table are very close, suggesting that the dataset's samples are relatively easy to distinguish. This also indicates that both linear and nonlinear

Table 2 Evaluation results for Z-Alizadeh Sani dataset dataset

Dataset	Data balance method	Classification method	Optimizer	Balancing ratio	Accuracy	Time (second)	Memory (MB)	CPU (%)	Fitness score
SET2	Quadratic	Ensemble learning	WOA	1.04	0.9886	0.61	9.23	24.7	0.6827
		Random forest	WOA	0.92	0.9879	3.22	9.5	25.4	0.6595
		SVM	PSO	1.04	0.9545	4.24	9.35	25	0.6387
Cubic	Cubic	Ensemble learning	PSO	1.06	0.9887	1.65	9.23	24.7	0.6785
		Random forest	Optuna	0.98	0.9883	1.06	9.46	25.3	0.6704
		SVM	Optuna	1.09	0.967	4.04	9.07	24.2	0.6613
Borderline-SMOTE	Borderline-SMOTE	Ensemble learning	Optuna	1	0.9612	2.03	9.75	26.1	0.6342
		Random forest	Optuna	1	0.9534	5.6	9.82	26.3	0.6108
		SVM	Optuna	1	0.9534	0.53	10.04	26.8	0.6225
SMOTE	SMOTE	Ensemble learning	PSO	0.97	0.9765	4.9	9.96	26.6	0.6242
		Random forest	PSO	0.93	0.9685	8.47	9.67	25.9	0.6163
		SVM	WOA	0.95	0.9453	11.07	9.86	26.4	0.6067
SVM-SMOTE	SVM-SMOTE	Ensemble learning	WOA	0.96	0.9687	1.47	9.75	26.1	0.6418
		Random forest	PSO	0.93	0.9606	1.38	9.57	25.7	0.6447
		SVM	Optuna	1	0.9302	0.78	9.82	26.3	0.6144
ADASYN	ADASYN	Ensemble learning	WOA	1	0.925	2.1	10.5	27.5	0.59
		Random forest	PSO	1	0.91	1.9	10.2	26.7	0.57
		SVM	Optuna	1	0.94	1.2	10.8	27.2	0.61

methods produce similar results, highlighting the effectiveness of different balancing approaches for this dataset.

7.4 Performance evaluation for stroke (SET4)

Stroke is a condition that can occur suddenly, making accurate and timely diagnosis critically important. It significantly impacts individuals' lives regardless of age. Stroke occurs due to a sudden disruption in the brain's blood supply, leading to permanent damage to body functions. If not addressed promptly, it not only lowers the patient's quality of life but also poses a severe risk to their survival. In this study, the Stroke dataset was selected, as with other datasets, due to its substantial impact on human life and its significant class imbalance, which effectively demonstrates the utility of balancing methods. The dataset has an imbalance ratio of 0.051, making it a particularly challenging case. Even the smallest improvement in stroke diagnosis can play a critical role in improving quality of life and reducing mortality rates. Detailed information about the dataset is provided in Sect. 2.1.4. In line with the primary objective of this study, optimization of different combinations was applied to this dataset as well. The results for this dataset, referred to as Set4, are presented in Table 4.

Set4 stands out as a dataset with significantly higher class imbalance compared to other datasets, making data balancing critically important. Under such conditions, learning from minority class samples becomes highly challenging. Table 4 shows that the best result is achieved using the combination of quadratic interpolation, WOA optimizer, and ensemble learning, with a balancing ratio of 1.04. The results indicate that accuracy improves significantly while the process is completed in a short time and with minimal resource requirements. Overall, these improvements in data balancing clearly contribute to enhancing the diagnosis of stroke, highlighting the importance of addressing class imbalance in such critical medical datasets.

7.5 Performance evaluation for kidney (SET5)

Kidney diseases are serious health conditions that can lead to life-threatening complications, where timely diagnosis and treatment are of critical importance. When the kidneys fail to function properly, toxic substances can accumulate in the body, adversely affecting the functions of other organs. If left untreated, this not only severely reduces the patient's quality of life but also puts their life at significant risk. As with the other datasets in this study, the Kidney Disease dataset was selected due to the substantial impact of this condition on human health and its significant class imbalance, which effectively demonstrates the utility of balancing methods. The dataset has an imbalance ratio of 0.6. Any improvement in the diagnosis of kidney diseases can play a vital role in enhancing patients' quality of life and reducing mortality rates. Detailed information about the dataset is provided in Sect. 2.1.5. In line with the primary objective of this study, the optimization of different combinations was applied to this dataset as well. The results for this dataset, referred to as Set5, are presented in Table 5.

Table 3 Evaluation results for Diabetes dataset

Dataset	Data balance method	Classification method	Optimizer	Balancing ratio	Accuracy	Time (second)	Memory (MB)	CPU (%)	Fitness score
SET3	Quadratic	Ensemble learning	PSO	1.06	0.9275	5.03	8	8	0.5573
		Random forest	PSO	1.07	0.9278	13.71	8	8	0.5348
		SVM	PSO	1.06	0.9082	4.15	8	8	0.5461
Cubic		Ensemble learning	WOA	1.06	0.9275	0.55	7.8	7.8	0.5721
		Random forest	WOA	1.09	0.92	3.35	8.3	8.3	0.5511
		SVM	WOA	1.06	0.8958	1.13	8	8	0.5453
Borderline-SMOTE		Ensemble learning	Optuna	1	0.9233	0.64	6.7	6.7	0.5598
		<i>Random forest</i>	<i>PSO</i>	<i>0.9</i>	<i>0.969</i>	<i>6.87</i>	<i>8.7</i>	<i>8.7</i>	<i>0.5707</i>
		SVM	WOA	1.04	0.9419	0.48	2.8	2.8	0.5925
SMOTE		Ensemble learning	Optuna	1	0.9466	6.62	13.2	13.2	0.4865
		Random forest	Optuna	1	0.97	1.93	13.2	13.2	0.5151
		SVM	PSO	0.9	0.8797	7.23	6.6	6.6	0.5396
SVM-SMOTE		Ensemble learning	Optuna	1	0.94	1.74	6.2	6.2	0.5423
		Random forest	PSO	0.93	0.9727	4.3	7.6	7.6	0.5444
		SVM	Optuna	1	0.86	1.46	6.9	6.9	0.5363
ADASYN		Ensemble learning	Optuna		0.88	2	7	7	0.52
		Random forest	PSO		0.89	5	8	8	0.53
		SVM	Optuna		0.85	2.5	7.5	7.5	0.51

Table 4 Evaluation Results for Stroke Dataset

Dataset	Data balance method	Classification method	Optimizer	Balancing ratio	Accuracy	Time (second)	Memory (MB)	CPU (%)	Fitness score
Set4	Quadratic	Ensemble learning	WOA	1.04	1	1.23	3.59	9.6	0.6089
		Random forest	Optuna	1.06	1	0.64	4.9	13.1	0.5392
Cubic		SVM	WOA	0.9	0.8157	3.89	2.32	6.2	0.5435
		Ensemble learning	Optuna	0.99	1	2.04	5.01	13.4	0.5885
		Random forest	PSO	<i>1</i>	<i>1</i>	<i>0.64</i>	<i>4</i>	<i>10.7</i>	<i>0.5877</i>
		SVM	PSO	0.9	0.8421	3.87	1.98	5.3	0.5803
Borderline-SMOTE		Ensemble learning	WOA	1.01	0.8571	1.57	3.59	9.6	0.5082
		Random forest	WOA	1.05	0.9145	18.3	3.93	10.4	0.4999
		SVM	WOA	1.05	0.7948	3.07	3.29	8.8	0.478
		Ensemble learning	WOA	1.05	0.8547	1.56	3.99	10.6	0.4856
SMOTE		Random forest	PSO	1.05	0.9145	20.38	2.71	7.2	0.5613
		SVM	WOA	1.05	0.7948	5.71	2.94	7.9	0.4916
		Ensemble learning	PSO	1.05	0.88	19.69	2.09	5.6	0.5713
		Random forest	Optuna	1	0.8833	6.06	2.47	6.6	0.5787
SVM-SMOTE		SVM	WOA	1.008	0.7833	0.72	3.55	9.5	0.5375
		Ensemble learning	PSO	1.03	0.85	20	2.5	6	0.56
		Random forest	Optuna	1.02	0.86	6.5	3	7	0.57
		SVM	WOA	1.01	0.78	1	4	10	0.53

The best result is marked in bold, the second-best in bolditalic, and the third-best in italic

Analyzing the balancing operations for the Kidney dataset in Table 5, it is evident that the best results are achieved using balancing methods based on quadratic and cubic interpolation. The top-performing combination is cubic interpolation combined with SVM and PSO, yielding the best results. The optimized balancing ratio was determined to be 0.94, meaning that increasing the minority class count from 150 to 235 (majority class count $\times$ balancing ratio) produced the most effective results. Upon examining the other outcomes, cubic interpolation, PSO, and SVM also delivered noteworthy results, highlighting the significant contribution of nonlinear methods to balancing operations. This underscores the effectiveness of nonlinear approaches in addressing class imbalance and enhancing classification performance.

7.6 Performance evaluation for WPBC (SET6)

Breast cancer is one of the most common cancers among women and is the most prevalent type of cancer overall. In this study, results for another breast cancer dataset, WPBC, are presented. Compared to the WDBC dataset, WPBC is more imbalanced, with an imbalance ratio of 0.31. This allows the study to evaluate the impact of the proposed method across different datasets for the same disease. Detailed information about the WPBC dataset is provided in Sect. 2.1.6. Table 6 presents the results for various combinations tested on the WPBC dataset. The results include both accuracy and resource consumption metrics, as the objective of this study is to optimize all these metrics simultaneously. In Table 6, the best result is highlighted in bold, the second-best in bolditalic, and the third-best in italic.

The objective of our study is to optimize both the quality and quantity of synthetic data generation. Specifically, in data balancing, the goal is to optimize not only the synthetic data but also classification accuracy and resource consumption. Upon reviewing Table 6, it is evident that for SET6, the combination of cubic interpolation for balancing, Optuna for optimizing the data balancing ratio, and SVM for classification achieves this goal more effectively. This combination results in an accuracy of 0.9672, completed in 2.23 s, with 2.62 MB memory usage and 7% CPU consumption. This indicates that the proposed method can operate efficiently on simple systems with fast processing times. Additionally, for this dataset, both Borderline-SMOTE and SVM-SMOTE balancing methods also yield successful results. This suggests that in SET6, where the minority class exhibits both nonlinear and linear distributions of examples, both linear and nonlinear methods can achieve high accuracy, demonstrating the adaptability of these methods under varying conditions.

7.7 Performance comparison for all sets

In this study, methods were tested on different datasets that are of vital importance to humanity. The primary goal of our method is to optimize the data balancing ratio, a factor that has often been overlooked in the literature, and to adapt cubic and quadratic interpolation methods for the data balancing problem to improve classification performance. Classification performance in this study takes into account both accuracy and resource consumption, because no matter how high the accuracy

Table 5 Evaluation results for kidney dataset

Dataset	Data balance method	Classification method	Optimizer	Balancing ratio	Accuracy	Time (second)	Memory (MB)	CPU (%)	Fitness score
SET5	Quadratic	Ensemble learning	PSO	1.02	1	0.4125	2.88	7.7	0.626
		Random forest	WOA	1.04	1	1.3	3.1	8.3	0.6112
Cubic		SVM	PSO	0.96	1	0.69	2.77	7.7	0.6295
		Ensemble learning	PSO	1.06	1	0.62	2.88	7.7	0.6249
		Random forest	PSO	0.9	1	2.42	3.14	8.4	0.6036
		SVM	PSO	0.94	1	1.43	2.58	6.9	0.6342
		Ensemble learning	Optuna	1	0.9666	1.03	2.95	7.9	0.5961
		Random forest	Optuna	1	0.98	1.5	4	10.7	0.554
Borderline-SMOTE		SVM	WOA	0.92	0.9727	1.84	3.89	10.4	0.5536
		Ensemble learning	Optuna	1	0.9733	0.28	3.07	8.2	0.5994
		Random forest	Optuna	1	0.98	1.47	4.15	11.1	0.5488
SMOTE		SVM	Optuna	1	0.96	2.4	2.99	8	0.5825
		Ensemble learning	WOA	0.97	1	2.25	1.57	4.3	0.6016
		Random forest	Optuna	1	0.9733	1.54	2.95	7.9	0.5981
		SVM	WOA	0.96	0.9731	6.69	3.6	9.8	0.5983
SVM-SMOTE		Ensemble learning	WOA	0.96	0.95	3	2	5	0.58
		Random forest	Optuna	0.95	0.92	2.5	3	7	0.57
ADASYN		SVM	WOA	0.94	0.91	6.5	4	8	0.55

The best result is marked in bold, the second-best in bolditalic, and the third-best in italic

Table 6 Evaluation results for WPBC dataset

Dataset	Data balance method	Classification method	Optimizer	Balancing ratio	Accuracy	Time (second)	Memory (MB)	CPU (%)	Fitness score
SET6	Quadratic	Ensemble learning	PSO	1.02	0.9672	5.99	3.63	9.7	0.6018
		Random forest	WOA	1.02	0.9836	11.7	3.51	9.4	0.5946
	Cubic	SVM	WOA	1.02	0.9836	7.78	3.25	8.7	0.6228
		Ensemble learning	Optuna	1.02	0.9502	4.89	3.85	10.3	0.5766
Borderline-SMOTE	Random forest	Random forest	Optuna	0.93	0.8818	6.5	4.41	11.48	0.4918
		SVM	Optuna	1.02	0.9672	2.23	2.62	7	0.6658
	Ensemble learning	Ensemble learning	PSO	0.9	0.9772	3.62	3.89	10.4	0.6018
		Random forest	WOA	0.99	0.98	1.85	4.6	12.3	0.5655
SMOTE	SVM	SVM	<i>Optuna</i>	<i>1</i>	<i>0.945</i>	<i>0.26</i>	<i>2.95</i>	<i>7.9</i>	<i>0.6388</i>
		Ensemble learning	Optuna	1	0.934	0.93	5.57	14.9	0.4844
	Random forest	Random forest	WOA	0.92	0.9775	5.38	4.82	12.9	0.5404
		SVM	PSO	0.98	0.99	2.2	4.71	12.6	0.5658
SVM-SMOTE	Ensemble learning	Ensemble learning	Optuna	1	0.94	1.74	4.64	12.4	0.5371
		Random forest	Optuna	1	0.9766	4.17	2.92	7.8	0.649
	SVM	SVM	Optuna	1	0.86	1.46	2.58	6.9	0.5965
		Ensemble learning	Optuna	1	0.92	2	10	14	0.52
ADASYN	Random forest	Random forest	Optuna	1	0.95	5	9	11	0.54
		SVM	Optuna	1	0.84	3.5	8	10	0.5

is, methods with excessive resource consumption will not be practical for use. The study includes a comprehensive analysis by combining different balancing methods, optimization methods, and classification techniques. This analysis utilizes metrics such as accuracy, precision, recall, F1-score, time, memory usage, and CPU usage. The results will demonstrate the impact of balancing methods on both accuracy and resource consumption, which is crucial for future research. Table 7 presents the results for each set, showing the best combinations across three different scenarios: the scenario where the balance ratio is optimized, the case of full balance where both the minority and majority classes have equal samples, and the results when no balancing method is applied. Only the combinations with the best results are displayed in the table, with the best result for each set highlighted in bold.

Table 7 presents the results for each dataset, showing the best combinations across three different scenarios. Upon general inspection, it is evident that the optimal balance ratios vary depending on the characteristics of the dataset. This is quite natural. In some datasets, where the distributions are more suitable for separation, the balance ratios remain low, while for datasets with more complex separations, a higher number of synthetic samples are required. On the other hand, it is observed that WOA and PSO optimization methods are prominent, with EL and SVM classifiers also performing very successfully. Notably, this success is achieved without a significant increase in resource consumption.

Upon examining the results, it is clear that the outcomes obtained with the optimized balance ratios offer both higher performance and lower resource consumption compared to fully balanced ratios. For example, in SET1 and SET2, using the quadratic interpolation method and optimizing the ratio with WOA, high accuracies of 0.9886 and 1 were achieved for the EL classifier. These results were obtained with lower time and memory usage compared to the fully balanced ratio outcomes. This demonstrates that it is not necessary to completely resolve class imbalance in datasets, and that determining the optimal balance ratio through proper optimization methods leads to more efficient results. In the analyses performed using SVM classifier on SET3 and SET5, Borderline-SMOTE and cubic interpolation methods stand out. Specifically, when the balance ratio was optimized with Borderline-SMOTE (at 1.04), an accuracy of 0.9419 was achieved for SVM, whereas with a fully balanced dataset, this value dropped to 0.8266. Similarly, for SET5, the optimized ratio using cubic interpolation with PSO yielded an accuracy of 1 at a ratio of 0.94. This demonstrates that optimized ratios provide superior performance not only in accuracy but also in other metrics (precision, recall, F1-score). In SET4 and SET6, while CPU and memory usage are low with fully balanced ratios, a noticeable drop in accuracy and other performance metrics is observed. For example, in SET6, with the optimized ratio using cubic interpolation and PSO, an accuracy of 0.9672 was achieved, compared to 0.846 with the fully balanced ratio. This indicates that fully balanced ratios negatively affect performance. Particularly in datasets with low imbalance ratios (e.g., 0.05), optimized balancing ratios lead to significant improvements in metrics such as accuracy and F1-score.

In conclusion, the overall success of the methods used in this study stands out due to the high accuracy, precision, and generalization capabilities achieved by determining the optimal balancing ratios and training classifiers with these ratios. The

Table 7 Analysis of the best combinations in different balancing scenarios

Dataset	Data balancing method	Balancing type	Balancing ratio	Classifier	Accuracy	Precision	Recall	F1-score	Time	Memory usage	CPU usage
SET1	Quadratic interpolation	Optimized ratio with WOA	1.04	Ensemble learning	0.9886	0.9729	1	0.9863	0.63	9.23	15.7
	Quadratic interpolation	Full balanced with WOA	1	Ensemble learning	0.9651	0.9444	0.971	0.9577	0.16	4.64	11.1
	No method	No balance	0.4	Ensemble learning	0.8347	0.8297	0.951	0.8863	0.11	3.089	15.3
SET2	Quadratic interpolation	Optimized ratio with WOA	0.93	Ensemble learning	1	1	1	1	1.67	9.35	15
	Quadratic interpolation	Full balanced with WOA	1	Ensemble learning	0.965	0.9634	0.976	0.969	0.17	4.28	14
SET3	No method	No balance	0.59	Ensemble learning	0.9766	0.9682	0.968	0.9682	0.14	3.16	12.7
	Borderline-SMOTE	Optimized ratio with WOA	1.04	SVM	0.9419	0.9277	0.875	0.9	0.48	2.8	2.8
	Borderline-SMOTE	Full balanced with WOA	1	SVM	0.8266	0.8036	0.868	0.8343	0.05	1.16	2.8
SET4	No method	No balance	0.53	SVM	0.8135	0.7176	0.763	0.7393	0.03	0.19	2.5
	Quadratic interpolation	Optimized ratio with WOA	1.04	Ensemble learning	1	1	1	1	1.23	3.59	9.6
	Quadratic interpolation	Full balanced with WOA	1	Ensemble learning	0.9987	0.9973	1	0.9986	4.85	3.73	2.3
SET5	No method	No balance	0.6	Ensemble learning	0.9418	0.9419	1	0.97	0.88	1.24	2.2
	Cubic interpolation	Optimized ratio with PSO	0.94	SVM	1	1	1	1	1.43	2.58	6.9
	Cubic interpolation	Full balanced with PSO	1	SVM	0.9866	0.9868	0.987	0.9868	0	2	5.7
No method	No balance		0.37	SVM	0.9833	0.9565	1	0.9777	0.01	1.8	5.2

Table 7 (continued)

Dataset	Data balancing method	Balancing type	Balancing ratio	Classifier	Accuracy	Precision	Recall	F1-score	Time	Memory usage	CPU usage
SET6	Cubic interpolation	Optimized ratio with PSO	1.04	SVM	0.9672	0.9339	1	0.9687	11	2.69	7.2
	Cubic interpolation	Full balanced with PSO	1	SVM	0.846	0.8666	0.83	0.8478	0.02	0.1	6
	No method	No balance	0.05	SVM	0.85	1	0.25	0.4	0.01	0.019	7

nonlinear cubic and quadratic interpolation methods, adapted for the first time in this field, have shown superiority over traditional methods, particularly in optimizing balance ratios and enhancing classification performance. The results demonstrate that these methods contribute not only to accuracy, precision, and F1-score but also do not lead to a significant increase in resource consumption, highlighting their advantage in both improving classification success and offering cost and resource efficiency. Moreover, when compared to fully balanced ratios, the positive impact of optimized balancing ratios on performance is clearly observed. While fully balanced ratios generally reduce resource consumption, they can also decrease classification performance. In contrast, datasets without balancing procedures showed severe negative impacts on classification performance due to class imbalance, emphasizing the critical role of balancing methods, especially the proposed nonlinear interpolation techniques, in datasets with class imbalance. The methods proposed in this study have led to significant improvements in the diagnosis of critical diseases such as stroke, diabetes, and kidney disease, which directly affect human life. The results of the study highlight the importance and value of the proposed approach, where even the smallest improvements can save a life. These methods, which can enhance classification performance even in datasets with low imbalance ratios, are expected to set a new standard in AI applications for health care. Therefore, this study demonstrates the importance of data balancing in applications where it is often overlooked and emphasizes how vital it is to ensure effective data balancing in such contexts.

The system on which the method is applied is crucial for its resource consumption and runtime. This study was run on a simple computer with an Intel i7-7300 CPU, 16 GB RAM, and 256 GB SSD running Windows. This demonstrates that the proposed method can run efficiently even on basic systems. This is particularly significant for low- and middle-income countries, where the high cost of health care makes it essential to develop software that can operate on simple systems. Such applications, which can reduce costs and increase accessibility, would play a crucial role in making healthcare solutions more affordable and widely adopted.

8 Discussion

Recently, with the widespread use of the internet and online records, massive amounts of data are being collected. As the volume of data increases, issues such as the degradation of standards, selecting the correct data for classification or other problems, and data organization arise. A major issue for classification methods is the imbalance in the distribution of examples within the data. In today's data-driven world, the idea of improving data quality has become inevitable. This study proposes some improvements to address data imbalance. A review of the literature reveals that there is insufficient research on mitigating data imbalance, with traditional balancing methods (such as SMOTE and its variants) commonly used. This is because most researchers opt to improve classification accuracy by focusing on enhancing classifiers, which is why new data balancing methods are rarely proposed. In this study, cubic and quadratic interpolation methods, frequently used in image resolution enhancement, are modified for generating one-dimensional data and their

contribution to this field is discussed. Another gap identified in the literature is the use of fixed values for the balancing ratio during dataset balancing. This study offers a new perspective by optimizing the balancing ratios using different optimization methods and discussing their impact on classification. It emphasizes the necessity of considering resource consumption in addition to classification accuracy, encouraging researchers to pursue studies in this direction. Furthermore, this study highlights the importance of data improvements in the context of the massive amounts of data available today. Future research will likely focus not only on improving classifiers but also on applying various improvements to the data itself.

9 Conclusion

The issue of imbalanced data distributions is particularly prevalent in the health-care domain, as it is often not feasible to collect balanced examples for different classes (e.g., sick vs. healthy individuals). As a result, this study focuses on dataset balancing and the optimization of balancing ratios using six different publicly available datasets with imbalanced examples: WDBC, WPBC, Z-Alizadeh Sani, Kidney, Diabetes, and Stroke. The common characteristic of these datasets is that they are related to widely recognized diseases and exhibit significant imbalance ratios. This study has three main objectives. First, it aims to enrich the field of dataset balancing by modifying methods, traditionally constrained in scope, with more diverse approaches. Second, unlike previous literature, it focuses on optimizing the dataset balancing ratio to determine how much synthetic data should be generated during balancing. Lastly, the study discusses not only the contribution to accuracy but also the effect of balancing on resource consumption. To achieve the first objective, cubic and quadratic interpolation methods, typically used in image processing, were adapted for dataset balancing. This introduces new balancing methods to complement SMOTE and its derivatives, and for the first time, the contribution of nonlinearity in interpolation methods is discussed in this context. For the second objective, three different optimization methods (PSO, WOA, and Optuna) were applied to optimize the balance ratios for each dataset. Additionally, evaluations were conducted using two different scenarios (full balance and imbalance), and the contribution of optimization to the balancing process was analyzed. After dataset balancing, the datasets were classified using three different classifiers (EL, RF, and SVM), and the impact of optimized balancing ratios on these classifiers was observed. A total of 54 different combinations were tested, allowing for a comprehensive analysis. In addition to metrics measuring quality such as accuracy, precision, recall, and F1-score, system metrics, such as runtime, memory usage, and CPU usage, were also calculated to demonstrate the practical applicability of the methods in simple systems. Since much of the literature focuses primarily on classification accuracy, this analysis is expected to make an impact on future research by highlighting the importance of data improvement.

Human health is one of the most valuable aspects of life. Any improvement made in this area is of great importance, as even the smallest contribution can save a life. Based on the analysis in this study, it is evident that the modified interpolation-based

methods produced very successful results for health care, and the optimization of dataset balancing improved classification accuracy while maintaining resource consumption at reasonable levels. Considering the system used for the study, it is concluded that the method is accessible to all sectors. Additionally, the analysis highlights the need for researchers to shift some of their focus from classifiers to data improvement, emphasizing the importance of addressing data imbalance in health-care applications.

Author contributions All authors contributed significantly to the conception, design, data acquisition, analysis, and interpretation of the study. The authors were involved in drafting the manuscript, and all authors critically revised it for important intellectual content.

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Data availability Data supporting this study is openly available from <https://archive.ics.uci.edu/dataset/45/heart+disease>), UCI Machine Learning Repository: <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>, <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>, <https://www.kaggle.com/datasets/mansoordaku/ckdisease>, <https://www.kaggle.com/code/anishsaman taray/wisconsin-wpbc-breast-cancer-prognosis>, <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>. No datasets were generated or analyzed during the current study.

Code availability Any custom code or software applications developed for this study are available from the corresponding author upon reasonable request. Detailed information on the algorithms or codes used is provided in the Methods section.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Ethical approval No ethical approval is required for the study.

Consent to participate All participants provided informed consent before participating in the study. They were made aware of the purpose of the research, potential risks, and their right to withdraw at any stage.

Consent for publication Participants were explicitly informed about the possible publication of the research outcomes. Any identifiable data or images have been anonymized or pseudonymized to protect the participants' privacy.

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References

- Hasib KMd, Iqbal MdS, Shah FM, Al Mahmud J, Popel MH, Showrov MdIH, Ahmed S, Rahman O (2020) A survey of methods for managing the classification and solution of data imbalance problem. *J Comput Sci* 16(11):1546–1557. <https://doi.org/10.3844/jccsp.2020.1546.1557>
- Narwane S, Sawarkar S (2019) Machine learning and class imbalance: a literature survey. *Ind Eng J*. <https://doi.org/10.26488/iej.12.10.1202>
- Malave N, Nimkar AV (2020) A survey on effects of class imbalance in data pre-processing stage of classification problem. *Int J Comput Syst Eng* 6(2):63. <https://doi.org/10.1504/ijcsyse.2020.111203>
- Yuan X, Chen S, Zhou H, Sun C, Yuwen L (2023) CHSMOTE: convex hull-based synthetic minority oversampling technique for alleviating the class imbalance problem. *Inf Sci* 623:324–341. <https://doi.org/10.1016/j.ins.2022.12.056>
- Yeung M, Sala E, Schönlieb C-B, Rundo L (2022) Unified Focal loss: generalising Dice and cross entropy-based losses to handle class imbalanced medical image segmentation. *Comput Med Imaging Graph* 95:102026. <https://doi.org/10.1016/j.compmedimag.2021.102026>
- Moradi M, Hamidzadeh J (2024) Handling class imbalance and overlap with a hesitation-based instance selection method. *Knowl-Based Syst* 294:111745–111745. <https://doi.org/10.1016/j.knosys.2024.111745>
- Rodríguez-Torres F, Martínez-Trinidad JF, Carrasco-Ochoa JA (2022) An oversampling method for class imbalance problems on large datasets. *Appl Sci* 12(7):3424. <https://doi.org/10.3390/app12073424>
- Ishaq A, Sadiq S, Umer M, Ullah S, Mirjalili S, Rupapara V, Nappi M (2021) Improving the prediction of heart failure patients' survival using SMOTE and effective data mining techniques. *IEEE Access* 9:39707–39716. <https://doi.org/10.1109/access.2021.3064084>
- Bellinger C, Sharma S, Japkowicz N, Zaïane OR (2019) Framework for extreme imbalance classification: SWIM—sampling with the majority class. *Knowl Inf Syst* 62(3):841–866. <https://doi.org/10.1007/s10115-019-01380-z>
- Hayashi T, Fujita H (2021) One-class ensemble classifier for data imbalance problems. *Appl Intell* 52(15):17073–17089. <https://doi.org/10.1007/s10489-021-02671-1>
- Sowjanya AM, Mrudula O (2022) Effective treatment of imbalanced datasets in health care using modified SMOTE coupled with stacked deep learning algorithms. *Appl Nanosci*. <https://doi.org/10.1007/s13204-021-02063-4>
- Tanapol K, Liu C, Hu R, Han S, Chen H, Lin Y (2023) A self-inspected adaptive SMOTE algorithm (SASMOTE) for highly imbalanced data classification in healthcare. *BioData Min* 16(1):15. <https://doi.org/10.1186/s13040-023-00330-4>
- Rajalakshmi R, Sivakumar P, Krishna Kumari L, Chengathir Selvi M (2023) A novel deep learning model for diabetes mellitus prediction in IoT-based healthcare environment with effective feature selection mechanism. *J Supercomput* 80:271–291. <https://doi.org/10.1007/s11227-023-05496-6>
- Lopo JA, Hartomo KD (2023) Evaluating sampling techniques for healthcare insurance fraud detection in imbalanced dataset. *J Ilm Tek Elektro Komput Dan Inform* 9(2):223–238
- Chowdhury MM, Ayon RS, Hossain MS (2024) An investigation of machine learning algorithms and data augmentation techniques for diabetes diagnosis using class imbalanced BRFSS dataset. *Healthcare Anal* 5:100297. <https://doi.org/10.1016/j.health.2023.100297>
- Elangovan VS, Devarajan R, Khalaf OI, Sharif MS, Elmedany W (2024) Analysing an imbalanced stroke prediction dataset using machine learning techniques. *Karbala Int J Mod Sci* 10(2):8. <https://doi.org/10.33640/2405-609x.3355>
- Mia R, Khanam S, Mahjabeen A, Ovy NH, Ghimire D, Park M, Begum MIA, Hosen ASMS (2024) Exploring machine learning for predicting cerebral stroke: a study in discovery. *Electronics* 13(4):686. <https://doi.org/10.3390/electronics13040686>
- Nizam-Ozogur H, Orman Z (2024) A heuristic-based hybrid sampling method using a combination of SMOTE and ENN for imbalanced health data. *Expert Syst*. <https://doi.org/10.1111/exsys.13596>
- Gholampour S (2024) Impact of nature of medical data on machine and deep learning for imbalanced datasets: clinical validity of SMOTE is questionable. *Mach Learn Knowl Extr* 6(2):827–841. <https://doi.org/10.3390/make6020039>

20. Ameksa M, Ellassad ZEA, Lamjadli S, Mousannif H (2024) Predicting stroke events with a proactive fusion system: a comprehensive study on imbalance class handling in computational biomechanics. *Comput Methods Biomech & Biomed Eng*. <https://doi.org/10.1080/10255842.2024.2363946>
21. Sun J, Li H, Fujita H, Fu B, Ai W (2020) Class-imbalanced dynamic financial distress prediction based on Adaboost-SVM ensemble combined with SMOTE and time weighting. *Inform Fus* 54:128–144. <https://doi.org/10.1016/j.inffus.2019.07.006>
22. Zhang R, Lu S, Yan B, Yu P, Tang X (2023) A density-based oversampling approach for class imbalance and data overlap. *Comput Ind Eng* 186:109747–109747. <https://doi.org/10.1016/j.cie.2023.109747>
23. Rendón E, Alejo R, Castorena C, Isidro-Ortega FJ, Granda-Gutiérrez EE (2020) Data sampling methods to deal with the big data multi-class imbalance problem. *Appl Sci* 10(4):1276. <https://doi.org/10.3390/app10041276>
24. Seiffert C, Khoshgoftaar TM, Van Hulse J, Napolitano A (2010) RUSBoost: a hybrid approach to alleviating class imbalance. *IEEE Trans Syst, Man, Cybern—Part A: Syst Hum* 40(1):185–197. <https://doi.org/10.1109/TSMCA.2009.2029559>
25. Li Z, Zheng T, Wang Y, Cao Z, Guo Z, Fu H (2021) A novel method for imbalanced fault diagnosis of rotating machinery based on generative adversarial networks. *IEEE Trans Instrum Meas* 70:1–17. <https://doi.org/10.1109/tim.2020.3009343>
26. Xiao C, Wang S (2021, December 5) An experimental study of class imbalance in federated learning. *ArXiv.org*. <https://doi.org/10.1109/SSCI50451.2021.9660072>
27. Davagdorj K, Lee JS, Pham VH, Ryu KH (2020) A comparative analysis of machine learning methods for class imbalance in a smoking cessation intervention. *Appl Sci* 10(9):3307. <https://doi.org/10.3390/app10093307>
28. Ganganwar V, Rajalakshmi R (2023) Employing synthetic data for addressing the class imbalance in aspect-based sentiment classification. *J Inform Telecommun* 8:167–188. <https://doi.org/10.1080/24751839.2023.2270824>
29. Razavi-Far R, Farajzadeh-Zanjani M, Wang B, Saif M, Chakrabarti S (2019) Imputation-based ensemble techniques for class imbalance learning. *IEEE Trans Knowl Data Eng* 33:1988–2001. <https://doi.org/10.1109/tkde.2019.2951556>
30. Sangapu SC, Prasad KSN, Kannan RJ, Chen TM, Sathiyarayanan M (2023) Impact of class imbalance in VeReMi dataset for misbehavior detection in autonomous vehicles. *Soft Comput*. <https://doi.org/10.1007/s00500-023-08003-4>
31. Attaallah A, Khan RA (2022) SMOTEDNN: a novel model for air pollution forecasting and AQI classification. *Comput, Mater & Continua (Print)* 71(1):1403–1425. <https://doi.org/10.32604/cmc.2022.021968>
32. Wolberg W, Mangasarian O, Street N, Street W (1995) Breast cancer Wisconsin (Diagnostic). UCI Mach Learn Repos. <https://doi.org/10.24432/C5DW2B>
33. Jeyasingh S, Veluchamy M (2017) Modified bat algorithm for feature selection with the Wisconsin diagnosis breast cancer (WDBC) dataset. *Asian Pacific J Cancer Prev: APJCP* 18(5):1257–1264. <https://doi.org/10.22034/APJCP.2017.18.5.1257>
34. Abdar M, Zomorodi-Moghadam M, Zhou X, Gururajan R, Tao X, Barua PD, Gururajan R (2020) A new nested ensemble technique for automated diagnosis of breast cancer. *Pattern Recogni Lett* 132:123–131. <https://doi.org/10.1016/j.patrec.2018.11.004>
35. Omondiagbe DA, Veeramani S, Sidhu AS (2019) Machine learning classification techniques for breast cancer diagnosis. In: *IOP Conference Series: Materials Science and Engineering*, vol. 495, p. 012033. <https://doi.org/10.1088/1757-899x/495/1/012033>
36. Wolberg W, Street W, Mangasarian O (1995) Breast cancer Wisconsin (Prognostic). UCI Mach Learn Repos. <https://doi.org/10.24432/C5GK50>
37. Kumar P, Nair GG (2021) An efficient classification framework for breast cancer using hyper parameter tuned random decision forest classifier and bayesian optimization. *Biomed Signal Process Control* 68:102682. <https://doi.org/10.1016/j.bspc.2021.102682>
38. Egwom OJ, Hassan M, Tanimu JJ, Hamada M, Ogar OM (2022) An LDA–SVM machine learning model for breast cancer classification. *BioMedInformatics* 2(3):345–358. <https://doi.org/10.3390/biomedinformatics2030022>
39. Saad Almutairi S, Manimurugan K, Majed Aborokbah B-G, Narmatha C (2023) Breast cancer classification using deep Q learning (DQL) and gorilla troops optimization (GTO). *Appl Soft Comput* 142:110292–110292. <https://doi.org/10.1016/j.asoc.2023.110292>

40. Alizadehsani R, Roshanzamir M, Sani Z (2017) Z-Alizadeh Sani. UCI Mach Learn Repos. <https://doi.org/10.24432/C5Q31T>
41. Gupta A, Kumar R, Arora HS, Raman B (2021) C-CADZ: computational intelligence system for coronary artery disease detection using Z-Alizadeh Sani dataset. *Appl Intell*. <https://doi.org/10.1007/s10489-021-02467-3>
42. Gupta A, Arora HS, Kumar R, Raman B (2021) DMHZ: a decision support system based on machine computational design for heart disease diagnosis using Z-Alizadeh sani dataset. In: 2021 International Conference on Information Networking (ICOIN). <https://doi.org/10.1109/icoin50884.2021.9333884>
43. Valarmathi R, Sheela T (2021) Heart disease prediction using hyper parameter optimization (HPO) tuning. *Biomed Signal Process Control* 70:103033. <https://doi.org/10.1016/j.bspc.2021.103033>
44. Rubini L, Soundarapandian P, Eswaran P (2015) Chronic kidney disease. UCI Mach Learn Repos. <https://doi.org/10.24432/C5G020>
45. Amirgaliyev Y, Shamilulu S, Serek A (2018, October 1) Analysis of chronic kidney disease dataset by applying machine learning methods. In: 2018 IEEE 12th International Conference on Application of Information and Communication Technologies (AICT). <https://doi.org/10.1109/ICAICT.2018.8747140>
46. Mahboob T, Ijaz A, Shahzad A, Kalsoom M (2018) Handling missing values in chronic kidney disease datasets using KNN. K-Means K-Medoids Algorithms. <https://doi.org/10.1109/icosst.2018.8632179>
47. Manonmani M, Balakrishnan S (2020) Feature selection using improved teaching learning based algorithm on chronic kidney disease dataset. *Proced Comput Sci* 171:1660–1669. <https://doi.org/10.1016/j.procs.2020.04.178>
48. Smith JW, Everhart JE, Dickson WC, Knowler WC, Johannes RS (1988) Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. In: *Proceedings of the Symposium on Computer Applications and Medical Care* (pp. 261–265). IEEE Computer Society Press.
49. Lakhwani K, Bhargava S, Hiran KK, Bunde MM, Somwanshi D (2020, December 1) Prediction of the onset of diabetes using artificial neural network and pima indians diabetes dataset. In: 2020 5th IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE). <https://doi.org/10.1109/ICRAIE51050.2020.9358308>
50. Shabtari MM, Kumar Shukla V, Singh H, Nanda I (2021, March 1) Analyzing PIMA Indian diabetes dataset through data mining tool “RapidMiner.” In: 2021 International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE). <https://doi.org/10.1109/ICACITE51222.2021.9404741>
51. Madan P, Singh V, Chaudhari V, Albagory Y, Dumka A, Singh R, Gehlot A, Rashid M, Alshamrani SS, AlGhamdi AS (2022) An optimization-based diabetes prediction model using CNN and Bi-directional LSTM in real-time environment. *Appl Sci* 12(8):3989. <https://doi.org/10.3390/app12083989>
52. Fedesoriano (2024) Stroke prediction dataset. Kaggle. <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>.
53. Santhiappan S, Ravindran B (2017) Class imbalance learning. *Adv Comput Commun*. <https://doi.org/10.34048/2017.1.f1>
54. Pes B (2020) Learning from high-dimensional biomedical datasets: the issue of class imbalance. *IEEE Access* 8:13527–13540. <https://doi.org/10.1109/access.2020.2966296>
55. Majzoub HA, Elgedawy I (2020) AB-SMOTE: an affirmative borderline SMOTE approach for imbalanced data binary classification. *Int J Mach Learn Comput* 10(1):31–37. <https://doi.org/10.18178/ijmlc.2020.10.1.894>
56. Hairani H, Saputro KE, Fadli S (2020) K-means-SMOTE for handling class imbalance in the classification of diabetes with C4.5, SVM, and naive Bayes. *J Teknol Dan Sist Komput* 8(2):89–93. <https://doi.org/10.14710/jtsiskom.8.2.2020.89-93>
57. Gyoten D, Ohkubo M, Nagata Y (2020) Imbalanced data classification procedure based on SMOTE. *Total Quality Science* 5(2):64–71. <https://doi.org/10.17929/tqs.5.64>
58. Satapathy SK, Mishra S, Mallick PK, Chae G-S (2021) ADASYN and ABC-optimized RBF convergence network for classification of electroencephalograph signal. *Pers Ubiquit Comput*. <https://doi.org/10.1007/s00779-021-01533-4>
59. Khan TM, Xu S, Khan ZG, Uzair chishti M (2021) Implementing multilabeling, ADASYN, and relief techniques for classification of breast cancer diagnostic through machine learning: efficient

- computer-aided diagnostic system. *J Healthcare Eng* 2021:1–15. <https://doi.org/10.1155/2021/5577636>
60. Alhudhaif A (2021) A novel multi-class imbalanced EEG signals classification based on the adaptive synthetic sampling (ADASYN) approach. *PeerJ Comput Sci* 7:e523. <https://doi.org/10.7717/peerj-cs.523>
 61. Ramadhan NG (2021) Comparative analysis of ADASYN-SVM and SMOTE-SVM methods on the detection of type 2 diabetes mellitus. *Sci J Inf* 8(2):276–282. <https://doi.org/10.15294/sji.v8i2.32484>
 62. Guo J, Wu H, Chen X, Lin W (2022) Adaptive kernel smote-svm algorithm for imbalanced data classification. *SSRN Electron J*. <https://doi.org/10.2139/ssrn.4139567>
 63. Sujitha R, Paramasivan B (2023) Optimal progressive classification study using SMOTE-SVM for stages of lung disease. *Automatika* 64(4):807–814. <https://doi.org/10.1080/00051144.2023.2218167>
 64. Smiti S, Soui M (2020) Bankruptcy prediction using deep learning approach based on borderline SMOTE. *Inf Syst Front*. <https://doi.org/10.1007/s10796-020-10031-6>
 65. Wang K-J, Adrian AM, Chen K-H, Wang K-M (2015) A hybrid classifier combining Borderline-SMOTE with AIRS algorithm for estimating brain metastasis from lung cancer: a case study in Taiwan. *Comput Methods Programs Biomed* 119(2):63–76. <https://doi.org/10.1016/j.cmpb.2015.03.003>
 66. Ning Q, Zhao X, Ma Z (2021) A novel method for Identification of Glutarylation sites combining Borderline-SMOTE with Tomek links technique in imbalanced data. *IEEE/ACM Trans Comput Biol Bioinf* 19:2632–2641. <https://doi.org/10.1109/tcbb.2021.3095482>
 67. Duan Q, Wang L, Twizell EH (2005) A new weighted rational cubic interpolation and its approximation. *Appl Math Comput* 168(2):990–1003. <https://doi.org/10.1016/j.amc.2004.09.041>
 68. Keys R (1981) Cubic convolution interpolation for digital image processing. *IEEE Trans Acoust Speech Signal Process* 29(6):1153–1160. <https://doi.org/10.1109/tassp.1981.1163711>
 69. Gruber B (1967) Numerical determination of the relative minimum of a function of several variables by quadratic interpolation. *Appl Math* 12(2):87–100. <https://doi.org/10.21136/am.1967.103074>
 70. Fan M (2011) Quadratic interpolation and some operator inequalities. *J Math Inequal* 3:413–427. <https://doi.org/10.7153/jmi-05-36>
 71. HU G (2009) Generalized genetic particle swarm optimization for supply chain optimization. *J Comput Appl* 28(11):2840–2843. <https://doi.org/10.3724/sp.j.1087.2008.02840>
 72. Gu H, Xu L (2011) Perceptive particle swarm optimization algorithm for constrained optimization problems. *J Comput Appl* 31(1):85–88. <https://doi.org/10.3724/sp.j.1087.2011.00085>
 73. Coello CAC, Pulido GT, Lechuga MS (2004) Handling multiple objectives with particle swarm optimization. *IEEE Trans Evol Comput* 8(3):256–279. <https://doi.org/10.1109/tevc.2004.826067>
 74. Nasiri J, Khiyabani FM (2018) A whale optimization algorithm (WOA) approach for clustering. *Cogent Math & Stat* 5(1):1483565. <https://doi.org/10.1080/25742558.2018.1483565>
 75. Qiu Y, Zhou J, Khandelwal M, Yang H, Yang P, Li C (2021) Performance evaluation of hybrid WOA-XGBoost, GWO-XGBoost and BO-XGBoost models to predict blast-induced ground vibration. *Eng Comput* 38:4145–4162. <https://doi.org/10.1007/s00366-021-01393-9>
 76. Chen W, Hong H, Liu B, Pradhan B, Wang Y, Shirzadi A, Pirasteh S, Alesheikh AA, Shahabi H, Panahi S, Rezaie F, Li S, Jaafari A, Pradhan B, Ahmad BB (2019) Spatial prediction of landslide susceptibility using GIS-based data mining techniques of ANFIS with whale optimization algorithm (WOA) and grey wolf optimizer (GWO). *Appl Sci* 9(18):3755–3755. <https://doi.org/10.3390/app9183755>
 77. Srinivas P, Katarya R (2022) hyOPTXg: OPTUNA hyper-parameter optimization framework for predicting cardiovascular disease using XGBoost. *Biomed Signal Process Control* 73:103456. <https://doi.org/10.1016/j.bspc.2021.103456>
 78. Mujilawati S, Sholihin M, Wardhani R (2021) Optimasi Hyperparameter TensorFlow dengan Menggunakan Optuna di Python: Study Kasus Klasifikasi Dokumen Abstrak Skripsi. *Jurnal Media Informatika Budidarma* 5(3):1084. <https://doi.org/10.30865/mib.v5i3.3090>
 79. Li J, Castagna J (2004) Support vector machine (SVM) pattern recognition to AVO classification. *Geophys Res Lett*. <https://doi.org/10.1029/2003gl018299>
 80. Gilmore S, Hofmann-Wellenhof R, Soyer HP (2010) A support vector machine for decision support in melanoma recognition. *Exp Dermatol* 19(9):830–835. <https://doi.org/10.1111/j.1600-0625.2010.01112.x>
 81. Zhang Z (2008) Research of multi-class algorithm based on fuzzy support vector machine. *J Comput Appl* 28(7):1681–1683. <https://doi.org/10.3724/sp.j.1087.2008.01681>

82. Mantas CJ, Castellano JG, Moral-García S, Abellán J (2018) A comparison of random forest based algorithms: random credal random forest versus oblique random forest. *Soft Comput*. <https://doi.org/10.1007/s00500-018-3628-5>
83. Survey on Random Forest Algorithm used in Mood analysis. (2018). *International Journal of Modern Trends in Engineering & Research*, 5(3):203–205. <https://doi.org/10.21884/ijmter.2018.5087.qrqzm>
84. Afanador NL, Smolinska A, Tran TN, Blanchet L (2016) Unsupervised random forest: a tutorial with case studies. *J Chemom* 30(5):232–241. <https://doi.org/10.1002/cem.2790>
85. SHI Y (2009) Ensemble learning application in improvement of particle swarm optimization algorithm. *J Comput Appl* 29(3):868–870. <https://doi.org/10.3724/sp.j.1087.2009.00868>
86. Zhang Y (2009) A new age estimation method based on ensemble learning. *Acta Autom Sinica* 34(8):997–1000. <https://doi.org/10.3724/sp.j.1004.2008.00997>
87. Zhang C-X, Zhang J-S (2011) A survey of selective ensemble learning algorithms. *Chinese J Comput* 34(8):1399–1410. <https://doi.org/10.3724/sp.j.1016.2011.01399>
88. Hospodsky D, Yamamoto N, Peccia J (2010) Accuracy, precision, and method detection limits of quantitative pcr for airborne bacteria and fungi. *Appl Environ Microbiol* 76(21):7004–7012. <https://doi.org/10.1128/aem.01240-10>

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